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Generative Digital Twin Methods for Spacecraft Telemetry Anomaly Detection: Toward Data-Driven FDIR Design and Testing

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Abstract

Spacecraft Fault Detection, Isolation and Recovery (FDIR) relies predominantly on model-based threshold logic. Such logic flags abrupt sensor faults reliably, yet it is structurally blind to a second kind of fault: the *contextual* anomaly, which alters the temporal pattern of a telemetry signal while its amplitude stays within the nominal range. In the NASA SMAP/MSL benchmark of real in-orbit telemetry, 43 of the 105 labelled anomalies are of this kind. This thesis asks whether generative models trained on real spacecraft telemetry can close that detection gap, and how far synthetic fault data can substitute for, augment, and characterise the data-driven detectors an operational FDIR system would deploy. The work proceeds in four phases. The detection baselines are evaluated on all 81 telemetry channels under the point-adjust protocol; the generative experiments use a clean channel-level held-out protocol, in which the generator never sees the channels reserved for evaluation. Data-driven detectors trained on real telemetry close the contextual gap by a wide margin, even from the primary telemetry channel alone: an unsupervised Variational Autoencoder and an LSTM regression network raise the mean F_1 on contextual channels from the classical baseline's 0.052 to 0.638 and 0.836 respectively at the oracle threshold, the LSTM reproducing the published benchmark to within 0.006. A conditional Wasserstein GAN, extended with four architectural contributions, synthesises fault windows that reproduce the contextual class to a Fréchet distance of 0.141 and match the real autocorrelation structure closely. On unseen held-out channels a classifier trained on synthetic data alone is not significantly worse than one trained on real data (0.474 against 0.631; Wilcoxon $p = 0.13$); with only 20 channels the test is low-powered, so this is the absence of a detectable difference rather than a demonstration of equivalence. Several results temper the picture. For the unsupervised detectors the deployable threshold is the parameter-free $k = 3$ rule, which synthetic-informed calibration does not beat; synthetic augmentation does not improve on cross-channel real data; and real fault windows do not transfer across channels. The principal contribution is a sensitivity-analysis methodology that turns the generator's controllability to account, mapping detection F_1 and latency across fault severity and duration on channels the generator never saw. Performance ranges from $F_1 = 0.255$ for the smallest, shortest faults to

0.845 for the largest and longest, and the surface shows that fault duration matters more than fault severity for detectability. What emerges is a partial FDIR design data sheet of a kind that fixed labelled data cannot supply. The work therefore positions generative Digital Twin methods less as a replacement for real data than as a controllable instrument for designing and verifying spacecraft fault management.

Keywords: spacecraft FDIR; anomaly detection; spacecraft telemetry; generative adversarial networks; synthetic data; variational autoencoder; digital twin

Abstract in lingua italiana

I sistemi di Identificazione, Isolamento e Recupero dei Guasti (FDIR) dei veicoli spaziali si basano prevalentemente su logiche a soglia di tipo model-based. Queste logiche rilevano in modo affidabile i guasti improvvisi dei sensori, ma sono strutturalmente cieche rispetto a un secondo tipo di guasto: l'anomalia *contestuale*, che altera l'andamento temporale di un segnale di telemetria mentre l'ampiezza resta entro l'intervallo nominale. Nel benchmark NASA SMAP/MSL di telemetria reale in orbita, 43 delle 105 anomalie etichettate sono di questo tipo. La tesi indaga se modelli generativi addestrati su telemetria spaziale reale possano colmare questa lacuna di rilevamento, e in quale misura i dati di guasto sintetici possano sostituire, aumentare e caratterizzare i rilevatori data-driven che un sistema FDIR operativo impiegherebbe. Il lavoro si articola in quattro fasi. I rilevatori di base sono valutati su tutti gli 81 canali di telemetria con il protocollo point-adjust; gli esperimenti generativi adottano invece un protocollo pulito di held-out a livello di canale, in cui il generatore non osserva mai i canali riservati alla valutazione. I rilevatori data-driven addestrati su telemetria reale colmano in larga misura la lacuna sulle anomalie contestuali, e lo fanno a partire dal solo canale di telemetria primario: un Variational Autoencoder non supervisionato e una rete di regressione LSTM portano l' F_1 medio sui canali contestuali dallo 0,052 della linea di base classica a 0,638 e 0,836 rispettivamente alla soglia oracle, con l'LSTM che riproduce il benchmark pubblicato entro 0,006. Una GAN di Wasserstein condizionale, estesa con quattro contributi architetturali, sintetizza finestre di guasto che riproducono la classe contestuale con una distanza di Fréchet di 0,141 e riproducono fedelmente la struttura di autocorrelazione reale. Su canali held-out mai osservati, un classificatore addestrato solo su dati sintetici non risulta significativamente peggiore di uno addestrato su dati reali (0,474 contro 0,631; Wilcoxon $p = 0,13$); con soli 20 canali il test ha potenza limitata, quindi si tratta dell'assenza di una differenza rilevabile e non di una dimostrazione di equivalenza. Diversi risultati ridimensionano il quadro. Per i rilevatori non supervisionati la soglia operativa è la regola $k = 3$, priva di parametri, che la calibrazione informata dai dati sintetici non supera; l'aumento sintetico non migliora i dati reali provenienti da altri canali; e le finestre di guasto reali non si trasferiscono tra canali. Il contributo principale è una metodologia di analisi di sensitività

che sfrutta la controllabilità del generatore per mappare l' F_1 e la latenza di rilevamento al variare della severità e della durata del guasto, su canali mai visti. Le prestazioni vanno da $F_1 = 0,255$ per i guasti più piccoli e brevi a $0,845$ per i più grandi e lunghi, e la superficie mostra che la durata del guasto conta più della severità nel determinarne la rilevabilità. Ne risulta una scheda di progetto FDIR parziale, di un tipo che i dati etichettati fissi non possono fornire. Il lavoro propone quindi i metodi generativi di Digital Twin non tanto come sostituto dei dati reali, quanto come uno strumento controllabile per la progettazione e la verifica della gestione dei guasti dei veicoli spaziali.

Parole chiave: FDIR satellitare; rilevamento di anomalie; telemetria spaziale; reti generative avversarie; dati sintetici; autoencoder variazionale; digital twin

Contents

Abstract	i
Abstract in lingua italiana	iii
Contents	v
Introduction	1
1 Background and State of the Art	7
1.1 Fault Detection, Isolation and Recovery in Spacecraft Systems	7
1.1.1 The role of FDIR	7
1.1.2 Classical model-based FDIR	8
1.1.3 Machine-learning approaches to FDIR	8
1.1.4 Open challenges and the motivation for generative approaches . . .	9
1.2 Digital Twins: Definitions, Maturity and Relevance to FDIR	10
1.2.1 The Digital Twin concept and its evolution	10
1.2.2 The DT maturity matrix	10
1.2.3 Model-Based Systems Engineering as a development methodology .	11
1.2.4 Digital Twins for predictive maintenance and fault diagnosis	11
1.2.5 Positioning the present research	11
1.3 Generative AI for Fault Diagnosis Across Industries	12
1.3.1 Motivation: data scarcity and class imbalance	12
1.3.2 Generative Adversarial Networks for fault-data augmentation	12
1.3.3 Variational Autoencoders for label-free anomaly detection	13
1.3.4 Transformer-based models and temporal self-attention	14
1.3.5 Diffusion models: an emerging frontier	14
1.3.6 Cross-domain evidence and the aerospace gap	15
1.4 Spacecraft Telemetry Anomaly Datasets	16
1.4.1 The data-scarcity problem in spacecraft FDIR	16

1.4.2	The NASA SMAP and MSL dataset	16
1.4.3	The ESA Anomaly Detection Benchmark	17
1.4.4	Comparison and selection rationale	17
1.4.5	The identified research gap	18
2	Methodology and Experimental Framework	19
2.1	Research Question and Objectives	19
2.2	The NASA SMAP/MSL Dataset	20
2.2.1	Structure and origin	20
2.2.2	Anomaly taxonomy: point and contextual faults	20
2.2.3	Dataset statistics and the class-imbalance problem	22
2.3	Evaluation Protocol and Metrics	22
2.3.1	Precision, recall and the F1 score	23
2.3.2	The point-adjust protocol	23
2.3.3	Threshold calibration: deployable and oracle	24
2.4	Common Experimental Conventions	24
2.5	Clean Held-Out Protocol for the Generative Experiments	25
2.5.1	The channel as the unit of partition	26
2.5.2	The split	26
2.6	The Four-Phase Methodology	27
2.7	Motivating Simulation: the Drift-Fault Detection Gap	28
3	Detection Baselines on Real Spacecraft Telemetry	31
3.1	Phase 1 — The Classical Threshold Baseline	31
3.2	Phase 2, Step A — Supervised Telemetry Classification	32
3.3	Phase 2, Step B — Variational Autoencoder	34
3.4	Phase 2, Step C — LSTM Regression	36
3.5	Complete Method Comparison	39
3.6	Discussion	41
4	Conditional Generative Modelling of Fault Data	43
4.1	Theoretical Background	43
4.1.1	Generative adversarial networks and their failure modes	43
4.1.2	Wasserstein GAN with gradient penalty	44
4.1.3	Conditional generation and stabilisation	44
4.2	Architectural Contributions	45
4.3	Model Architecture and Training Protocol	46
4.4	Training Convergence	47

4.5	Generation Quality Evaluation	49
4.6	Train-on-Synthetic, Test-on-Real Validation	51
4.7	Discussion	52
5	Synthetic-to-Real Transfer and Sensitivity Analysis	55
5.1	Campaign 1 — Synthetic-to-Real Substitution	55
5.2	Campaign 2 — The Marginal Value of Data	57
5.2.1	Data efficiency: real windows from known channels do not transfer .	57
5.2.2	Augmentation: synthetic substitutes for, but does not augment, real data	59
5.3	Campaign 3 — Label-Free Threshold Calibration	60
5.3.1	The LSTM synthMedian result is a grid-resolution artefact	61
5.4	Campaign 4 — Sensitivity Analysis with a Controllable Fault Generator .	62
5.4.1	Injection methodology	63
5.4.2	The detection-performance surface	63
5.4.3	The detection-latency surface	65
5.4.4	Per-class behaviour	66
5.4.5	Robustness of the sensitivity surface	67
5.4.6	Channel P-1 as a genuine unseen-channel result	67
5.5	Statistical Summary	68
5.6	Discussion	69
6	Conclusions and Future Developments	71
6.1	Summary of Findings	71
6.2	Answer to the Research Question	72
6.3	Limitations	73
6.4	Future Developments	74
6.5	Closing Remark	75
	Bibliography	77
A	Implementation Details and Hyperparameters	81
A.1	Supervised Classifier (Phase 2, Step A)	81
A.2	Variational Autoencoder (Phase 2, Step B)	82
A.3	LSTM Regression Detector (Phase 2, Step C)	83
A.4	Conditional WGAN-GP (Phase 3)	83

B Per-Channel Detection Results	85
C Per-Class and Per-Channel Sensitivity Results	91
C.1 Detection performance by anomaly class	91
C.2 Detection latency by anomaly class	92
C.3 Per-channel detection at representative operating points	93
D Declaration on the Use of AI Tools	97
List of Figures	99
List of Tables	103
List of Symbols and Acronyms	107
Acknowledgements	109

Introduction

Spacecraft operate in an environment that offers no second chances. A fault left undetected during a communication blackout, or one that develops too slowly to trip a conventional alarm, can compromise a mission before a ground operator knows anything is wrong. The onboard logic that guards against this, grouped under the heading of Fault Detection, Isolation and Recovery (FDIR), is among the most safety-critical software a satellite carries. It is also expected to act more and more on its own, as missions move toward large constellations, on-orbit servicing, and deep-space operation where ground contact is intermittent at best. The need is sharpest for the Attitude and Orbit Control System. Empirical data from small-satellite missions attribute roughly 17.75% of the failures that contribute to system losses in the first thirty days of operation to attitude-control faults, which makes attitude control one of the largest single contributors to early-mission failure [3].

The FDIR paradigm dominant in flight today is model-based. A software replica of the spacecraft runs alongside the real system, residuals between predicted and measured behaviour are formed, and a fault is declared when a residual exceeds a manually calibrated threshold. This is fast, interpretable, frugal with onboard resources, and reliable on abrupt sensor faults [3]. But it has a structural blind spot. A fault that accumulates gradually, or that changes the temporal pattern of a signal without pushing its amplitude past the nominal range, produces no threshold crossing and is never raised. Such *contextual* faults are not a rare edge case: in the public record of real in-orbit anomalies studied here [15] they account for 43 of the 105 labelled events. The running-variance detector standard in classical FDIR misses them however the threshold is tuned, because the fault leaves no quantity for the threshold to act on.

This thesis asks whether that gap can be closed with data rather than with more hand-tuned rules. It then asks a harder question that the first one exposes. Scarcity of real fault data has long held data-driven FDIR back, so can generative models trained on the faults a spacecraft *has* seen produce enough realistic synthetic faults to train, calibrate, and characterise the detectors that must catch the ones it has not?

The data-driven turn and its unexamined assumption

The limitations of threshold logic have driven a steady move toward data-driven FDIR, in which a classifier learns a fault’s statistical signature directly from data instead of relying on a hand-specified residual test. The closest precedent for the present work is the study of machine-learning FDIR for spacecraft attitude control by Crotti and Colagrossi [4], who benchmarked Support Vector Machines and Multilayer Perceptrons against the classical variance detector across six canonical fault types. Their results are encouraging. The classifiers reach near-perfect diagnostic accuracy on known fault types, and a network monitoring filter-estimated angular rates detects actuator drift faults that the classical detector never flags. Their recommended architecture is complementary rather than competitive: classical detection handles fast, reliable flagging of obvious faults, while a machine-learning layer adds coverage of the failure modes that threshold logic cannot reach.

One assumption runs through that line of work, and it is the one this thesis sets out to test. Every classifier is trained only on data from a numerical simulator. However faithful that simulator, it injects faults according to predefined parametric models. Real spacecraft faults are different: they arise from physical degradation, thermal cycling, radiation damage, and manufacturing variability, processes whose statistical signatures need not match the simulated ones. The robustness of a simulator-trained classifier to the fault distribution it would actually meet in orbit is therefore unknown. And the severe class imbalance of real telemetry, where a fault may occupy a fraction of a percent of a channel’s lifetime, is not addressed by simulation-only training at all.

Two developments now make it possible to confront that assumption directly, for the first time. The first is the public release of real in-orbit telemetry with expert-verified anomaly labels: the NASA Soil Moisture Active Passive (SMAP) and Mars Science Laboratory (MSL) benchmark of Hundman et al. [15], and the more recent European Space Agency Anomaly Detection Benchmark of Kotowski et al. [21]. The second is the maturation of generative modelling for time series. Across manufacturing, energy, and process-control domains, generative augmentation has repeatedly improved downstream fault classification on imbalanced datasets; adversarial models supply targeted generation of specific fault types, and variational models supply unsupervised detection from nominal data alone [16, 18, 23, 25]. To the best of the candidate’s knowledge, no published work has brought the two together. Generative augmentation has not been applied to spacecraft telemetry fault detection on the SMAP/MSL or ESA benchmarks with the explicit aim of improving FDIR over a simulation-trained baseline. That gap defines the contribution

of this thesis.

Research question

The work is organised around a single research question:

Can generative models — specifically a Variational Autoencoder for unsupervised anomaly detection and a conditional Generative Adversarial Network for fault-data synthesis — trained on real spacecraft telemetry from the NASA SMAP/MSL benchmark, improve fault-detection accuracy and robustness relative to classical model-based FDIR, and to what extent can synthetic fault data substitute for, augment, and characterise the detectors that an operational FDIR system would deploy?

The question is deliberately bounded: it fixes the dataset, the architectures, and the comparison baselines. It is evaluable, because the benchmark supplies expert ground-truth labels and an established protocol against which every result can be measured [15]. And it is original, because no prior published work addresses this combination of generative modelling, real spacecraft telemetry, and the FDIR-improvement objective.

Approach

The investigation proceeds in four phases. Each builds on the quantitative results of the last, and all are evaluated on the NASA SMAP/MSL benchmark under the point-adjust protocol of Hundman et al. [15], so that results compare directly across the programme. The detection baselines use all 81 unique telemetry channels, once a duplicate annotation is resolved. The generative experiments use a clean channel-level held-out split and evaluate only on channels the generator never observed, so that the synthetic-to-real claims concern genuinely unseen faults rather than a retrospective benchmark.

The first phase establishes the baseline: a classical z-score threshold detector is evaluated across all channels, and a controlled attitude-dynamics simulation reproduces, in a spacecraft-engineering setting, the structural failure of variance-based detection on a slowly accumulating drift fault. The second phase builds the data-driven detectors on real telemetry, namely a supervised SVM/MLP classifier, an unsupervised Variational Autoencoder trained only on nominal data, and an LSTM regression detector whose agreement with the published benchmark of Hundman et al. [15] independently validates the experimental pipeline. The third phase introduces the first generative component: a conditional Wasserstein GAN with gradient penalty, trained on the real anomaly windows

of the training-channel split, with the realism of its output assessed by distributional metrics and a train-on-synthetic, test-on-real protocol on the held-out channels. The fourth phase asks what that synthetic data is worth in operational terms. It is tested as a substitute for real labelled training data, as an augmentation of it, as a label-free calibration signal, and finally as the input to a controlled sensitivity analysis that maps detector performance across fault severity and duration in a way no fixed labelled dataset can support.

Contributions

The thesis makes the following contributions to the literature on spacecraft FDIR and generative modelling for fault diagnosis.

1. It evaluates two unsupervised detectors on the NASA SMAP/MSL benchmark under the point-adjust protocol, a reconstruction-based Variational Autoencoder and a prediction-based LSTM regression network, and disaggregates their performance by the point/contextual fault taxonomy of spacecraft FDIR. The result is that data-driven detection trained on real spacecraft telemetry closes the contextual-anomaly gap left open by classical FDIR. Detectors of this kind have been studied on the benchmark before: the prediction-error LSTM is the method of Hundman et al. [15] themselves, and multivariate detectors such as those of Su et al. [27] and Audibert et al. [2] have since been evaluated on it. The contribution here is therefore the analysis through the FDIR fault-class lens, the use of these detectors as a controlled baseline for the generative experiments that follow, and an independent reproduction of the published LSTM benchmark that validates the pipeline to within 0.006.
2. It develops what is, to the best of the candidate's knowledge, the first conditional Wasserstein GAN trained on the real labelled anomaly windows of the NASA SMAP/MSL benchmark for FDIR fault-data synthesis and characterisation. Generative modelling of time series is itself well established; recurrent and adversarial generators such as those of Esteban et al. [6], Yoon et al. [33] and Geiger et al. [8] are direct precedents. The novelty claimed here is thus the target domain and objective, not the use of a GAN for sequence generation. Four architectural choices go beyond the standard formulation: a one-dimensional convolutional generator and critic, an explicit autocorrelation-matching loss, per-class specialised generators, and a spectrally normalised critic compatible with the gradient penalty. The output is validated both distributionally and through a downstream detection task.
3. Under a clean channel-level held-out protocol, it quantifies the operational value of

synthetic spacecraft fault data. On previously unseen channels a supervised detector trained on synthetic faults alone is not statistically worse than one trained on real faults; a paired test over the 20 held-out channels detects no difference, though with so few channels it is correspondingly low-powered. Synthetic augmentation, by contrast, does not improve on cross-channel real data, and synthetic-informed threshold calibration does not beat the classical default. Together with these negative findings, the results mark out where synthetic fault data helps an operational detector and where it does not.

4. It introduces a sensitivity-analysis methodology built on the controllability of the generative model. Synthetic faults are injected into clean nominal telemetry at controlled severity and duration, and the detector's F_1 and detection latency are mapped as two-dimensional surfaces. The output is an FDIR design data sheet, stating the minimum detectable fault and the detection latency across the operational parameter range. A fixed labelled dataset has no analogue for this, because in it each fault occurs at a single uncontrollable combination of magnitude and duration.

Together, these contributions move spacecraft FDIR from its present state of the art, machine-learning classifiers trained on simulated data alongside classical detection, toward the Cognitive Digital Twin envisaged in the maturity framework of Wei et al. [31]: a system that learns and continually characterises its own fault-detection capability from real operational data.

Structure of the thesis

The remainder of the manuscript is organised as follows. Chapter 1 reviews the state of the art in spacecraft FDIR, the Digital Twin maturity framework that positions the work, generative modelling for fault diagnosis across industries, and the spacecraft telemetry anomaly datasets, closing on the specific research gap addressed here. Chapter 2 formalises the research question, describes the dataset and the point-adjust evaluation protocol, sets out the four-phase methodology, and presents the attitude-dynamics simulation that frames the detection problem in spacecraft-engineering terms. Chapter 3 reports the detection baselines on real telemetry: the classical threshold detector and the supervised and unsupervised machine-learning detectors of Phases 1 and 2. Chapter 4 develops the conditional Wasserstein GAN of Phase 3 and evaluates the realism of its synthetic fault data. Chapter 5 reports the synthetic-to-real transfer experiments and the sensitivity analysis of Phase 4. Finally, Chapter 6 draws the conclusions, states the oper-

ational recommendation that follows from the results, and outlines directions for future development.

1 | Background and State of the Art

This chapter establishes the technical and scientific context of the thesis. Section 1.1 reviews Fault Detection, Isolation and Recovery (FDIR) in spacecraft systems, from the classical model-based approach in flight today to the recent machine-learning extensions, and identifies the limitation that motivates the present work. Section 1.2 introduces the Digital Twin maturity framework that situates the contribution within a broader systems-engineering trajectory. Section 1.3 surveys generative modelling for fault diagnosis across industrial domains, from which the methodological tools of this thesis are drawn. Section 1.4 reviews the spacecraft telemetry anomaly datasets that make a real-data study possible, and closes by stating the specific research gap addressed in the chapters that follow.

1.1. Fault Detection, Isolation and Recovery in Spacecraft Systems

1.1.1. The role of FDIR

The management of onboard health and operational status is a fundamental challenge in spacecraft design. As satellites become more autonomous and more mission-critical, detecting, isolating and recovering from hardware and software faults without ground intervention becomes essential. Fault Detection, Isolation and Recovery (FDIR) systems provide exactly this: automated supervision of the spacecraft's behaviour under unexpected conditions and malfunctions.

The need for FDIR is particularly acute for the Attitude and Orbit Control System (AOCS), among the subsystems most responsible for mission success and most vulnerable to failure. Empirical data from small-satellite missions show that roughly 17.75% of all failures contributing to system losses within the first thirty days of operation are

attributable to AOCS faults, which makes the AOCS one of the largest single contributors to early-mission failure [3]. As small-satellite platforms take on more complex objectives, including large constellations, in-orbit inspection and distributed sensing, the demand for reliable autonomous FDIR grows with them.

1.1.2. Classical model-based FDIR

The traditional approach to FDIR relies on model-based methods that exploit analytical redundancy. Instead of adding redundant physical hardware, a software model of the system runs in parallel with real operations. The outputs of real sensors and actuators are compared continuously against the model's predictions, residuals are computed and monitored, and a fault is flagged when a residual exceeds a defined threshold.

Colagrossi and Lavagna [3] present a complete fault-tolerant Attitude and Orbit Determination System (AODS) for small-satellite platforms that exemplifies this approach. It operates on raw measurements from gyroscopes, magnetometers, Sun sensors and GNSS receivers, passing them through a pipeline of low-pass filtering, reference-frame rotation, measurement correction and component-level fault detection. Detection at the component level uses the running variance and running mean over a sliding window of recent samples. The architecture handles six canonical fault types: spike, erratic, drift, hardover (or bias), data-loss and stuck faults [3, 17].

This classical architecture performs reliably under the tight computational constraints of small satellites. One limitation is nonetheless built in: detection sensitivity is governed entirely by manually calibrated variance thresholds, which must be tuned per sensor, per mission and per operational mode. Drift faults expose a deeper, structural weakness. A slow drift that does not reverse produces no variance signature, and so escapes detection altogether.

1.1.3. Machine-learning approaches to FDIR

The limitations of threshold-based detection motivate data-driven alternatives. Rather than requiring explicit models of nominal behaviour, machine-learning classifiers learn statistical representations of signal patterns directly from data, and can in principle capture fault signatures that resist simple threshold rules.

Crotti and Colagrossi [4] conduct a systematic investigation of machine-learning FDIR for spacecraft ADCS using a high-fidelity functional engineering simulator. They benchmark Support Vector Machines with linear, quadratic, cubic and Gaussian kernels, and

Multilayer Perceptron neural networks of varying depth and width, against the classical variance detector across the six fault types above. The picture that emerges is nuanced. For four-class detection that mimics the classical detector’s output flags, the classifiers reach accuracies between 80% and 97%, comparable to but not uniformly above the classical detector’s consistent 93.75%; the classical detector, however, fails entirely on slowly evolving drift faults. For fault *diagnosis*, where the task is to identify the precise fault type given a known-faulty signal, accuracy approaches 100% across nearly all sensor–mode combinations. And a neural network trained on filter-estimated angular rates detects actuator drift faults almost immediately after onset, on a channel where the classical detector never triggers at all.

The main conclusion of Crotti and Colagrossi [4] is that machine-learning classifiers are best deployed not as replacements for classical methods but as a complementary diagnostic layer. Classical detection still provides fast, reliable flagging of obvious faults; the learned classifier adds higher-level diagnosis and coverage of failure modes that threshold logic cannot reach.

1.1.4. Open challenges and the motivation for generative approaches

A critical limitation runs through the machine-learning FDIR literature: the classifiers are trained almost exclusively on data from a numerical simulator. However accurate, a simulator injects fault signatures according to predefined parametric models, whereas real spacecraft faults arise from physical degradation, thermal cycling, radiation damage and manufacturing variability, phenomena whose statistical signatures may differ substantially from the simulated ones. The classifiers of Crotti and Colagrossi [4] have never seen real spacecraft telemetry, and their robustness to out-of-distribution fault signatures from actual on-orbit anomalies is unknown.

At the same time, real spacecraft telemetry with labelled anomalies has recently become publicly available: the NASA SMAP/MSL dataset of Hundman et al. [15] and the ESA Anomaly Detection Benchmark of Kotowski et al. [21]. This opens the possibility of training generative models that learn the true statistical distribution of real fault signals. Such models could then synthesise training data that reflects in-orbit fault behaviour more faithfully, and so improve classifier robustness beyond what simulation-only training can achieve. That possibility is the point of departure for this thesis.

1.2. Digital Twins: Definitions, Maturity and Relevance to FDIR

1.2.1. The Digital Twin concept and its evolution

The Digital Twin (DT) was first introduced by Grieves [11] in the context of product lifecycle management, described as comprising three elements: the physical product, its virtual counterpart, and the bidirectional connections between them. The concept was applied to aerospace by NASA, which defined a Digital Twin as a multi-physics, multi-scale, probabilistic, ultra-fidelity simulation that reflects, in a timely manner, the state of a corresponding physical twin on the basis of historical data, real-time sensor data and physical models [9].

In the two decades since, the concept has expanded considerably. Tao et al. [28] proposed a five-dimensional framework encompassing the physical component, the virtual counterpart, their interconnections, data and services. More recent work distinguishes three related but meaningfully different entities. A *Digital Model* is a virtual representation of a physical entity with no automated data exchange. A *Digital Shadow* adds automated data flow from the physical entity to the digital representation. A *Digital Twin* proper adds the reverse flow as well, from the digital entity back to the physical system, which lets the virtual model prescribe, reconfigure or control the physical twin autonomously.

1.2.2. The DT maturity matrix

Wei et al. [31] formalise this progression through a six-level maturity matrix. Level 1 (Descriptive) is a static digital model with no automated data flow. Level 2 (Analytical) enriches the model with simulation and analytical algorithms. Level 3 (Operational) establishes automated physical-to-digital data flow, transforming the model into a Digital Shadow. Level 4 (Prescriptive) adds automated digital-to-physical data flow, completing the bidirectional link that defines a true Digital Twin. Level 5 (Cognitive) integrates advanced artificial-intelligence and data-science algorithms for autonomous decision-making. Level 6 (Connected Cognitive) connects the twin to other digital systems and other twins, enabling system-of-systems reasoning.

This matrix maps directly onto the present work. A spacecraft FDIR system using classical model-based detection corresponds roughly to Level 3. The machine-learning extension of Crotti and Colagrossi [4] begins to approach Level 4. A system that also employs generative models trained on real telemetry, continuously updating its statistical picture of

fault behaviour from operational data, corresponds to Level 5: a Cognitive Digital Twin able to learn and adapt its fault-detection capability without human retraining.

1.2.3. Model-Based Systems Engineering as a development methodology

The development of high-maturity Digital Twins is substantially helped by Model-Based Systems Engineering (MBSE). Wei et al. [31] propose the DEVOTION methodology, a step-wise MBSE-based approach to developing twins from any starting maturity level to any target level. Applied to an electrical and electronics case study for space launch vehicles, DEVOTION shows how automated code generation from formal models can produce the communication interfaces needed for physical-to-digital and digital-to-physical data flows, and how probabilistic model checking, specifically Discrete-Time Markov Chain analysis, can be folded into the twin loop to give continuous confidence estimates for onboard components.

1.2.4. Digital Twins for predictive maintenance and fault diagnosis

Beyond spacecraft, integrating artificial intelligence with Digital Twins for predictive maintenance has attracted broad interest under the Industry 4.0 and 5.0 paradigms. Mikołajewska et al. [23] review generative-AI approaches to twin-based fault diagnosis and identify a central challenge common to all domains: industrial systems rarely generate enough labelled fault data for direct supervised learning, because catastrophic or rare faults occur infrequently by design. Generative models address this by learning the statistical structure of the available data and synthesising additional examples consistent with it. The review singles out four architectures of primary relevance, namely Generative Adversarial Networks, Variational Autoencoders, Transformer-based models and diffusion models. It reports that GANs typically improve fault-classification accuracy by 5–15%, that VAEs reach anomaly-detection precision of 85–95% by learning a compact representation of normal behaviour, and that Transformer-based models outperform recurrent architectures by 10–20% on long time series.

1.2.5. Positioning the present research

The maturity framework of Wei et al. [31] and the generative-AI review of Mikołajewska et al. [23] together define the conceptual space of this thesis. It is worth being precise

about scope. This thesis does not implement a Digital Twin in the strict sense: there is no synchronised virtual replica of a specific physical spacecraft, no automated physical-to-digital or digital-to-physical data flow, and no closed-loop reconfiguration of an operating system. What it develops are the generative *methods* that would form the learning and fault-management layer of a Cognitive (Level 5) Digital Twin for spacecraft FDIR: unsupervised detectors that learn from nominal telemetry, a controllable fault generator, and a procedure for characterising detector capability. In such a twin, generative models trained on real telemetry would be the engine that lets it improve its fault-detection capability beyond what simulation alone affords. The title’s phrase “generative Digital Twin methods” should be read in exactly this sense, as methods directed *toward* a Cognitive Digital Twin; the maturity framework is used throughout to place the contribution on a trajectory, not to claim that a twin is delivered. This positioning carries three concrete requirements: the generative model must be trained on real spacecraft telemetry, its output must be usable as training data for downstream FDIR detectors, and its architecture must stay within the onboard computational constraints of small satellites.

1.3. Generative AI for Fault Diagnosis Across Industries

1.3.1. Motivation: data scarcity and class imbalance

The application of generative artificial intelligence to fault diagnosis is one of the most active frontiers in industrial machine learning, and its basic motivation is data scarcity. Real fault events are rare by design in well-engineered systems, so labelled fault examples are systematically underrepresented relative to nominal operation. The imbalance is most severe in safety-critical domains such as aerospace, where a catastrophic fault may occur only once in a system’s lifetime, yet a detector must recognise it reliably from the first occurrence. Generative models address this in two ways. They can synthesise additional examples that reflect the learned fault distribution; and, when trained only on nominal data, they can act directly as anomaly detectors, flagging any signal they cannot faithfully reproduce, whether or not its specific fault type was seen during training.

1.3.2. Generative Adversarial Networks for fault-data augmentation

Generative Adversarial Networks, introduced by Goodfellow et al. [10], consist of two networks trained in opposition: a generator that produces synthetic samples and a dis-

criminator that tries to tell them from real ones. Through iterative adversarial training, the generator learns to produce samples whose statistical properties are hard to distinguish from the training data. Shao et al. [25] established the use of GANs for mechanical fault diagnosis, showing that an auxiliary-classifier GAN could learn from mechanical sensor signals and generate realistic one-dimensional raw data for rotating machinery. Their study set the template most later work follows: train on real fault data, generate synthetic minority-class examples, combine them with real data, and evaluate the downstream classifier against a real-only baseline.

A key practical difficulty in applying standard GANs to time-series fault data is training instability. The Wasserstein GAN with Gradient Penalty (WGAN-GP) of Gulrajani et al. [12] tackles both mode collapse and vanishing gradients by replacing the discriminator loss with a Wasserstein-distance estimate enforced through a gradient-penalty term, and it has become the dominant variant in the fault-diagnosis literature. Jin et al. [18], for instance, applied a conditional WGAN-GP to generate specific fault categories of signals from sucker-rod pumping systems under severe class imbalance. The consistent finding across domains is that GAN-based augmentation improves downstream accuracy by 5–15% when the minority fault class is fewer than 5% of the available examples [23]. That condition applies directly to the spacecraft context, where some anomaly types in the SMAP/MSL dataset occupy fewer than 2% of total time steps.

A parallel line of work addresses the generation of realistic *time series* directly, rather than static feature vectors: recurrent conditional GANs for real-valued sequences [6], the convolutional waveform-synthesis approach of WaveGAN [5], the embedding-space approach of TimeGAN [33], and, for the anomaly-detection setting specifically, TadGAN [8], which couples adversarial reconstruction with a cycle-consistency objective and is evaluated on time-series anomaly benchmarks. These establish that adversarial models can capture temporal structure, but none target spacecraft telemetry or the SMAP/MSL benchmark, and the difficulty of enforcing temporal-correlation fidelity in particular motivates the explicit autocorrelation-matching supervision introduced in Chapter 4.

1.3.3. Variational Autoencoders for label-free anomaly detection

Variational Autoencoders, introduced by Kingma and Welling [20], take a fundamentally different approach that is especially valuable when labelled fault data is unavailable. A VAE encodes input data into a compressed latent representation governed by a learned probability distribution, then decodes samples from that distribution back into the origi-

nal space. The training objective combines a reconstruction loss with a regularisation term that keeps the latent distribution close to a standard normal prior, producing a smooth, structured latent space of normal operating conditions. Detection then exploits the reconstruction error. A test sample lying outside the distribution learned on nominal data cannot be reconstructed faithfully, and the resulting error exceeds a threshold calibrated on held-out nominal data, with no labelled fault examples required at all. Jakubowski et al. [16] applied VAE-based detection to industrial equipment degradation, catching incipient faults through a steadily rising reconstruction error well before physical failure. Fährmann et al. [7] proposed a lightweight LSTM-VAE for multi-channel industrial control systems, aimed explicitly at resource-constrained deployment, a constraint shared by small spacecraft.

1.3.4. Transformer-based models and temporal self-attention

The Transformer architecture of Vaswani et al. [30] has become a dominant approach for time-series modelling. Its self-attention mechanism relates any two time steps directly, regardless of how far apart they are, sidestepping the vanishing-gradient limitation that affects recurrent architectures on long sequences. Xu et al. [32] introduced the Anomaly Transformer for unsupervised time-series anomaly detection. The idea is that normal patterns show strong associations between adjacent points, while anomalies disrupt that local structure and create atypical long-range associations. By modelling a prior-association with a local Gaussian kernel alongside a learned series-association, and maximising their divergence through a minimax strategy, the Anomaly Transformer produces a principled anomaly score without labelled faults. It was evaluated on six benchmarks, among them the NASA SMAP and MSL datasets targeted by this thesis, where it achieves state-of-the-art results and so serves here as the upper-bound reference.

1.3.5. Diffusion models: an emerging frontier

Diffusion models generate data by reversing a gradual noise-addition process, and have overtaken GANs as the state of the art for high-dimensional data. Tashiro et al. [29] showed with CSDI that diffusion models conditioned on partial observations can generate high-fidelity multivariate time series, and Lopez Alcaraz and Strodthoff [22] showed that diffusion combined with structured state-space backbones outperforms GAN and VAE approaches on standard imputation and forecasting benchmarks. The main limitation in the present context is computational cost, though Song et al. [26] cut the required sampling steps to as few as ten to twenty with denoising diffusion implicit models. For off-line training-data augmentation, the primary application here since generation happens

on ground infrastructure, that overhead is acceptable. Diffusion modelling is therefore identified as a direction for future work rather than a component of the present implementation.

1.3.6. Cross-domain evidence and the aerospace gap

The convergence of evidence across manufacturing, energy, transportation and process control supports several conclusions. Generative augmentation consistently improves downstream classifier performance on imbalanced fault datasets, with GANs supplying targeted, fault-type-specific augmentation and VAEs supplying robust unsupervised detection. Transformer-based architectures outperform recurrent models on long, complex time series, a property that directly characterises spacecraft telemetry, and they have been validated on the exact datasets targeted here. And the pairing of a generative model for augmentation with a trained classifier for fault identification, the synergistic design recommended by Crotti and Colagrossi [4], is the dominant pattern in the industrial literature.

Against this backdrop, the spacecraft domain shows a clear gap. The overwhelming majority of machine-learning FDIR work in aerospace relies on simulator-generated training data, and no published work has applied GAN-, VAE- or diffusion-based augmentation to spacecraft telemetry fault classification using the NASA SMAP/MSL or ESA benchmark datasets. Table 1.1 summarises the four architectures reviewed above and their role in this thesis.

Architecture	Primary use in this thesis	Key advantage / limitation
Conditional GAN (WGAN-GP)	Synthesis of labelled fault windows for augmentation and transfer	Targeted per-class generation; requires labelled fault examples
Variational Autoencoder	Unsupervised anomaly detection from nominal data	No labelled faults needed; limited to the learned normal distribution
Anomaly Transformer	Upper-bound reference on the SMAP/MSL benchmark	State of the art; higher computational cost
Diffusion model	Future work	Highest generation fidelity; high sampling cost

Table 1.1: Generative architectures reviewed in Section 1.3 and their role in this thesis.

1.4. Spacecraft Telemetry Anomaly Datasets

1.4.1. The data-scarcity problem in spacecraft FDIR

A recurring obstacle to machine-learning FDIR for spacecraft has been the historical absence of publicly available real telemetry with verified anomaly labels. The field has lacked the standardised benchmarks that the industrial fault-diagnosis community relies on for objective comparison. Simulator-generated data, as used by Crotti and Colagrossi [4], gives a controlled and reproducible training environment, but it cannot replicate the statistical properties of real in-orbit fault events. Two landmark releases changed this: the NASA SMAP/MSL dataset of Hundman et al. [15] and the ESA Anomaly Detection Benchmark of Kotowski et al. [21], which together form the first public collections of real spacecraft telemetry with expert-verified anomaly labels.

1.4.2. The NASA SMAP and MSL dataset

The NASA SMAP/MSL dataset was published by Hundman et al. [15] alongside their LSTM-based detection method with nonparametric dynamic thresholding. It comprises

real telemetry from two operational NASA spacecraft: the Soil Moisture Active Passive (SMAP) satellite, launched in January 2015 to measure global soil moisture from low Earth orbit, and the Mars Science Laboratory (MSL, the *Curiosity* rover), which has explored the surface of Mars since August 2012. The dataset contains 82 labelled telemetry channels, 55 from SMAP and 27 from MSL, each a continuous time series of a sensor or actuator output recorded during real operations. The anomaly labels were assigned by NASA spacecraft operations engineers who identified genuine fault events in the historical record; they are neither injected nor simulated. Of the 105 labelled anomaly windows, 62 are point anomalies and 43 are contextual anomalies. The training split of each channel contains nominal data only, while the test split contains the mixture of nominal and anomalous operation against which detection is evaluated. Since its release the benchmark has become a standard testbed for time-series anomaly detection, evaluated by reconstruction- and prediction-based methods including OmniAnomaly [27], USAD [2] and the Anomaly Transformer [32]; this body of results provides the external reference points against which the detectors of Chapter 3 are positioned. The detailed per-channel statistics, which quantify the severe class imbalance that motivates the generative approach, are presented in Chapter 2 alongside the experimental protocol.

1.4.3. The ESA Anomaly Detection Benchmark

The ESA Anomaly Detection Benchmark (ESA-ADB) of Kotowski et al. [21] is a substantially larger and more complex resource, comprising annotated real-life telemetry from three ESA missions. Its annotations were produced by operations engineers at the European Space Operations Centre together with machine-learning experts, then cross-verified using standard anomaly-detection algorithms; this two-stage process yields more reliable ground truth than a single-reviewer or fully automated benchmark. The scale is considerable. The number of data points exceeds 700 million per mission, with more than 7 gigabytes of compressed data per mission, and the dataset preserves the practical difficulties of operational telemetry: irregular timestamps, varying sampling rates, communication gaps, and anomalies present in the training data itself.

1.4.4. Comparison and selection rationale

The NASA SMAP/MSL dataset is the primary experimental dataset of this thesis. Its manageable scale, clean train-test split, well-documented file structure, and existing benchmark results from Hundman et al. [15] and Xu et al. [32] make it the right basis for developing and validating the methodology. The ESA-ADB serves as the larger-scale validation target toward which the methodology would later be extended; its superior la-

bel quality and operational realism make it the more demanding and industrially relevant benchmark, but its scale and preprocessing complexity put it beyond the present scope. Between them, the two datasets define a research programme that starts with a tractable, well-benchmarked problem on real spacecraft telemetry and scales toward the operational complexity of European spacecraft monitoring.

1.4.5. The identified research gap

The literature reviewed in this chapter converges on a single, clearly defined gap. Classical model-based FDIR reliably detects abrupt sensor faults but is structurally incapable of detecting slowly evolving or contextual anomalies. Machine-learning classifiers trained on simulator data partially address this, but their robustness to real in-orbit fault distributions is untested, and the class imbalance of real telemetry is left unaddressed by simulation-only training. The generative-AI literature offers tools for both problems at once, yet no published work has applied generative augmentation to spacecraft telemetry fault detection on the SMAP/MSL or ESA benchmarks with the explicit aim of improving FDIR over a simulation-trained baseline. Closing that gap, by training generative models on real spacecraft telemetry and quantifying their value for FDIR design and testing, is the contribution of this thesis; the methodology that does so is set out in Chapter 2.

2 | Methodology and Experimental Framework

This chapter sets out the methodology that the experimental chapters implement. Section 2.1 states the research question and the objectives that decompose it. Section 2.2 describes the NASA SMAP/MSL dataset and quantifies the class imbalance that motivates the generative approach. Section 2.3 defines the evaluation protocol and metrics used uniformly across every phase, and Section 2.4 fixes the preprocessing and reproducibility conventions shared by all experiments. Section 2.6 presents the four-phase methodology as a whole. Finally, Section 2.7 reports a controlled attitude-dynamics simulation that reproduces, in a spacecraft-engineering setting, the detection gap that the remainder of the thesis sets out to close.

2.1. Research Question and Objectives

The thesis is organised around the research question stated in the Introduction, repeated here for reference:

Can generative models — specifically a Variational Autoencoder for unsupervised anomaly detection and a conditional Generative Adversarial Network for fault-data synthesis — trained on real spacecraft telemetry from the NASA SMAP/MSL benchmark, improve fault-detection accuracy and robustness relative to classical model-based FDIR, and to what extent can synthetic fault data substitute for, augment, and characterise the detectors that an operational FDIR system would deploy?

The question decomposes into four objectives, each addressed by one phase of the methodology. The first is to establish a quantitative detection baseline on real telemetry, and to confirm in a controlled simulation that classical variance-based FDIR fails structurally on the contextual fault class. The second is to determine whether machine-learning detectors trained on real telemetry close that gap, and by how much. The third is to train

a generative model on the real fault record and check that its synthetic output is statistically and temporally faithful to the real fault distribution. The fourth is to quantify what that synthetic data is worth: as a substitute for and augmentation of real labelled training data, and as the basis for a controlled sensitivity analysis of detector capability. All four are evaluable because the benchmark provides expert ground-truth labels and an established evaluation protocol, described in Section 2.3.

2.2. The NASA SMAP/MSL Dataset

2.2.1. Structure and origin

All experiments in this thesis use the NASA SMAP/MSL anomaly detection dataset of Hundman et al. [15]. It comprises real telemetry recorded from two operational NASA spacecraft: the Soil Moisture Active Passive (SMAP) Earth-observation satellite and the Mars Science Laboratory (MSL) *Curiosity* rover. There are 82 labelled telemetry channels, 55 from SMAP and 27 from MSL, each a single continuous time series of a sensor or actuator output. For every channel the data are split into a training portion, which by NASA’s annotation contains nominal operation only, and a test portion, which contains a mixture of nominal and anomalous operation. Each channel is stored as a two-dimensional array whose first column is the primary telemetry measurement and whose remaining columns encode command and context information. In line with the prior phases of this work, only the primary measurement (column 0) is used; extending the analysis to the full multivariate input is identified as future work in Chapter 6.

A practical detail of the ground-truth file deserves mention because it affects the channel count reported throughout. The annotation file `labeled_anomalies.csv` lists the channel identifier P-2 twice, with slightly different anomaly-window boundaries. This known quirk of the original dataset is resolved consistently in every experiment by retaining the first occurrence, which yields 81 unique channels for aggregate evaluation. Results are reported over these 81 channels unless a published comparison value requires the original 82.

2.2.2. Anomaly taxonomy: point and contextual faults

The ground truth labels each anomaly window as one of two classes. A *point* anomaly is a sudden amplitude excursion: the signal leaves its normal range abruptly and conspicuously. A *contextual* anomaly is a change in the temporal pattern of the signal while its amplitude stays within the normal range. The signal still looks superficially normal, but its oscillation rhythm, correlation structure or local dynamics have shifted. Of the 105

labelled anomaly windows, 62 are point anomalies and 43 are contextual.

This distinction is central to the thesis. The two classes correspond closely to the fault families of the spacecraft FDIR taxonomy of Colagrossi and Lavagna [3]: point anomalies cover the spike and hardover signatures defined by amplitude exceedance, while contextual anomalies cover the drift and erratic signatures defined by distributional shift within normal bounds. It is the contextual class that classical variance-based detection cannot reach, because the fault produces no amplitude quantity for a threshold to act on. Figure 2.1 illustrates the contextual class on SMAP channel P-1, the primary illustrative case used throughout this thesis: its three labelled anomaly windows stay entirely within the normal amplitude range, yet show a measurable change in oscillation pattern relative to nominal operation.

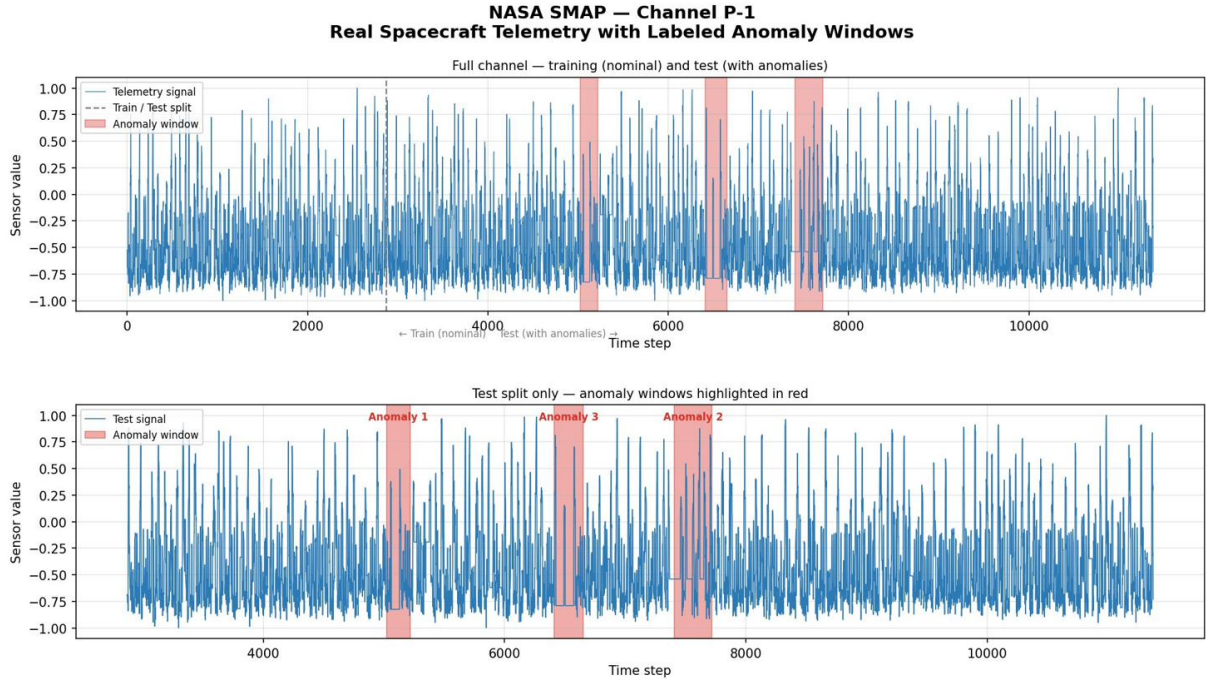


Figure 2.1: SMAP channel P-1 telemetry with its three labelled contextual anomaly windows (shaded). The signal remains within its normal amplitude range of approximately $[-1, +1]$ throughout the anomaly windows, yet its oscillation pattern and statistical distribution change relative to the nominal behaviour learned during training. This is the category of fault that classical variance-based threshold detection cannot reach, and that motivates the distributional and predictive detectors developed in Chapter 3.

2.2.3. Dataset statistics and the class-imbalance problem

The statistics computed directly from the ground-truth labels, summarised in Table 2.1, quantify the severity of the class imbalance that the generative augmentation of this thesis is designed to address. These descriptive statistics are computed over all 82 annotation rows of `labeled_anomalies.csv`, including the duplicated P-2 entry, and so describe the published dataset as distributed; the aggregate detection results in the following chapters are instead reported over the 81 unique channels, as explained in Section 2.2.1. The minimum per-channel anomaly rate is only 0.39%, and 39 of the 82 channels, 48% of the dataset, have anomaly rates below 5% of their total recorded time steps. With 43 of the 105 anomaly windows in the contextual class, the dataset combines two compounding difficulties: anomalies are rare, and a large fraction of them are of the type classical detection cannot register at all.

Property	Value
Total telemetry channels	82
SMAP channels	55
MSL channels	27
Total labelled anomaly windows	105
Point anomaly windows	62
Contextual anomaly windows	43
Minimum anomaly rate (per channel)	0.39%
Maximum anomaly rate (per channel)	50.07%
Mean anomaly rate (per channel)	12.28%
Median anomaly rate (per channel)	6.02%
Channels with anomaly rate below 5%	39 of 82 (48%)
Channels with anomaly rate below 10%	51 of 82 (62%)

Table 2.1: NASA SMAP/MSL dataset statistics computed from `labeled_anomalies.csv` [15], over all 82 annotation rows (including the duplicated P-2). Aggregate detection results in later chapters use the 81 unique channels.

2.3. Evaluation Protocol and Metrics

2.3.1. Precision, recall and the F_1 score

Detection performance throughout the thesis is reported using three metrics consistent with Hundman et al. [15] and the protocol of Kotowski et al. [21]. Writing TP , FP and FN for the counts of true-positive, false-positive and false-negative detections, precision is the fraction of flagged units that are true anomalies, recall is the fraction of true anomalies that are flagged, and the F_1 score is their harmonic mean:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = \frac{2 \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (2.1)$$

The F_1 score is the primary comparison metric, because it balances the competing objectives of catching anomalies (recall) and not raising spurious alarms (precision). The per-channel F_1 distribution is right-skewed: a few high-anomaly-rate channels are easy, while many low-rate channels are hard. Both the mean and the median F_1 across channels are therefore reported wherever the distinction is informative, following the practice of Xu et al. [32].

2.3.2. The point-adjust protocol

Detection is performed at the level of the individual time step, but evaluated under the point-adjust protocol introduced by Hundman et al. [15], which is the standard for this benchmark. Under point-adjust, for every contiguous ground-truth anomaly segment, if the detector flags *any* single time step within that segment, the entire segment is counted as correctly detected. The protocol expresses the operational requirement that an FDIR system must trigger at least once within a fault event to be considered successful; it does not require the detector to identify the precise onset and offset of the anomaly to the nearest sample.

It is worth being explicit about a known property of this protocol. As Kim et al. [19] demonstrate, point-adjust can substantially *inflate* the apparent F_1 on datasets with long anomaly segments, because a single correctly flagged time step is credited with the whole segment; absolute point-adjust scores are therefore not directly comparable to scores from stricter, point-wise protocols. Two considerations justify its use here. First, it is the protocol under which the reference benchmark of Hundman et al. [15] and the comparison results of Xu et al. [32] are reported, so adopting it is what makes the pipeline-validation comparison in Chapter 3 a genuine like-for-like one. Second, the protocol is applied identically to every method in this thesis. It raises the absolute level of all scores, but it is unbiased between detectors, so the *relative* ordering of methods, on which the

argument of the thesis rests, is unaffected. The absolute F_1 values reported throughout should accordingly be read as benchmark-comparable figures under point-adjust, not as the scores the same detectors would obtain under a point-wise protocol. The supervised classifier of Chapter 3 is instead evaluated at the window level, because the window is the natural unit of its prediction; this matches the convention used when that classifier is trained on synthetic data in Chapter 5.

2.3.3. Threshold calibration: deployable and oracle

The unsupervised detectors of this thesis, the Variational Autoencoder and the LSTM regression network, do not output a class label but a continuous anomaly score, and detection requires a threshold above which the score is taken to indicate a fault. Setting this threshold is the central practical difficulty of deployable unsupervised detection, because the real fault labels that would let it be optimised are exactly what is unavailable in operational FDIR. Two threshold strategies are therefore reported throughout. The deployable threshold is the classical k -sigma rule,

$$\theta_k = \mu_E + k \sigma_E, \quad (2.2)$$

where μ_E and σ_E are the mean and standard deviation of the detector’s error scores on its own nominal training data; the value $k = 3$ is used as the operational default, following the three-sigma convention, because it can be computed from training data alone. The best (or oracle) threshold is obtained by searching a grid of k values and training-error percentiles and selecting the one that maximises F_1 on the test set. The oracle threshold requires knowledge of the test labels and is therefore not deployable; it is reported only as an upper bound on what each architecture could achieve with perfect calibration. Throughout the thesis the $k = 3$ figure is the fair, deployable comparison, while the oracle figure quantifies the headroom a better label-free calibration rule could in principle recover, a question taken up directly in Chapter 5.

2.4. Common Experimental Conventions

To ensure that results compare directly across all four phases, every experiment shares a single set of preprocessing and reproducibility conventions.

Each channel’s training and test signals are normalised by the channel-specific mean and standard deviation computed on the training split, producing signals in z-score units. Where a representation in the $[-1, +1]$ range is required, for the generator output of

Chapter 4 and the features of the supervised classifier, the z-scores are clipped at $\pm 3\sigma$ and divided by three,

$$\tilde{x}_t = \text{clip}\left(\frac{x_t - \mu_{\text{train}}}{\sigma_{\text{train}}}, -3, +3\right) / 3, \quad (2.3)$$

so that the $\pm 3\sigma$ band, which contains 99.7% of nominal values, maps to $[-1, +1]$ while the amplitude semantics that distinguish the two anomaly classes are preserved.

The signal is processed in overlapping windows of 64 time steps with a step of 16, giving 75% overlap between consecutive windows. For window-level labelling, a window is marked anomalous if at least 10% of its time steps fall inside a ground-truth anomaly segment; this deliberately permissive overlap threshold ensures that boundary windows of short anomaly segments are captured as positive examples. All randomised operations use a fixed seed of 42. These conventions, a window of 64, a step of 16, an overlap threshold of 0.1, channel-level z-score normalisation, and the fixed seed, are held constant in every experiment, so that differences in performance are attributable to method rather than to preprocessing.

2.5. Clean Held-Out Protocol for the Generative Experiments

The detection baselines of Phases 1 and 2 are evaluated on all 81 channels in the standard way, using the train–test split that ships with the dataset: each detector is trained on a channel’s nominal training portion and evaluated on its test portion, and no anomaly label is ever used in training. The generative experiments of Phases 3 and 4 require a stricter protocol, for a reason that must be stated explicitly because it governs how the results may be interpreted.

The conditional generator of Chapter 4 is trained on real anomaly windows, and the only part of the dataset that carries anomaly labels is the test split. If the generator were trained on anomaly windows drawn from the same channels on which the synthetic-to-real detectors are later evaluated, the synthetic fault distribution would be shaped, indirectly through the generator, by the very anomalies later used to measure detection performance. Results obtained that way could be reported only as a retrospective benchmark on *seen* anomalies; they could not support the stronger claim that synthetic faults stand in for the *unseen* faults an operational detector must catch. This is, in precise terms, a train–test leakage problem: anomaly information from the evaluation channels would reach the detectors indirectly, through a generator that had been trained on it. An earlier version of these experiments in this work partitioned the data exactly that way, and the optimistic

results it produced are reported and withdrawn in Section 5.2.1 and Section 5.2.2. To make the stronger, generalisation claim admissible, Phases 3 and 4 are therefore run under a clean held-out protocol in which the channels used for evaluation are never seen by the generator at any stage.

2.5.1. The channel as the unit of partition

The natural first idea, holding out a fraction of each channel’s anomaly windows, is infeasible on this dataset. Of the 81 channels, 63 contain only a single labelled anomaly sequence, so holding out that one sequence would leave the channel with nothing to test on and nothing to calibrate the threshold against at the same time; a further 13 channels have two sequences and 5 have three, still too few for a stable within-channel split. The anomaly sequence is therefore not a viable unit of partition here.

Because the generator is trained on a pool of anomaly windows aggregated across many channels, the natural clean unit is instead the entire channel. The protocol holds out a set of complete channels: their anomalies never enter the generator’s training pool, never enter any detector’s training set, and are used only for the final synthetic-to-real evaluation. This is both stricter and more informative than a within-channel split, because it answers a genuine generalisation question: whether a generator that has never seen a given channel’s faults can still produce synthetic data that detects them.

2.5.2. The split

The 81 channels are partitioned, stratified by their primary anomaly class (point, contextual, or mixed) and fixed by the random seed 42, into 61 training channels and 20 held-out channels, with approximately one quarter of each class held out. The held-out set comprises 13 channels whose primary class is point, 6 contextual, and one mixed channel (C-1, labelled point and contextual). To keep the class-disaggregated held-out results internally consistent, the mixed channel C-1 is grouped with the *point* class in the per-class breakdowns of Chapter 4, Chapter 5 and Chapter C, so that the held-out set is reported as 14 point and 6 contextual channels whenever those results are split by class. The 20 held-out channels are A-8, B-1, C-1, D-14, D-4, D-5, D-9, E-10, E-8, F-3, G-1, G-2, G-3, G-4, M-2, P-1, P-2, R-1, T-2, and T-5. Channel P-1, the contextual channel used as the illustrative case throughout the thesis, falls in the held-out set, so its Phase 4 result becomes a genuine unseen-channel result, which is exactly the most demanding way to present it.

The split is computed once, in a single shared module imported by both the Phase 3

generator script and the Phase 4 detector scripts, so that the partition is identical across stages and cannot drift between them: the generator’s window-extraction loop skips the held-out channels, and every Phase 4 detector loop skips any channel not in the held-out set. All Phase 3 and Phase 4 results reported in this thesis use a generator trained under this protocol; the Phase 4 detector results and the train-on-synthetic transfer test are evaluated on the 20 held-out channels, while the Phase 3 distributional quality metrics are computed in-sample against the generator’s training windows. Every synthetic-versus-real comparison is tested for statistical significance with a paired Wilcoxon signed-rank test over the 20 held-out channels.

2.6. The Four-Phase Methodology

The investigation is structured in four sequential phases. Phases 1 and 2 are evaluated on all 81 channels under the protocol of Section 2.3; Phases 3 and 4 are evaluated under the clean held-out protocol of Section 2.5.

Phase 1 — Baseline. A classical z-score threshold detector is evaluated across all channels to establish the quantitative detection floor that subsequent methods must exceed, and the attitude-dynamics simulation of Section 2.7 confirms, in a controlled spacecraft setting, that variance-based detection fails structurally on a slowly accumulating drift fault.

Phase 2 — Data-driven detection on real telemetry. Three detectors are built and evaluated on real telemetry: a supervised SVM/MLP classifier operating on hand-crafted temporal features, an unsupervised Variational Autoencoder trained only on nominal data and scoring anomalies by reconstruction error, and an LSTM regression network scoring anomalies by one-step-ahead prediction error. The LSTM additionally validates the experimental pipeline by reproducing the published benchmark of Hundman et al. [15].

Phase 3 — Generative modelling of fault data. A conditional Wasserstein GAN with gradient penalty is trained on the real anomaly windows of the 61 training channels, conditioned on the point/contextual class label. The realism of its synthetic output is then evaluated in two complementary ways: distributional metrics that compare the synthetic windows against the generator’s own training windows (an in-sample fidelity check), and a train-on-synthetic, test-on-real protocol evaluated on the held-out channels.

Phase 4 — Synthetic-to-real transfer and sensitivity analysis. On the 20 held-out channels, the synthetic data is evaluated as a substitute for real labelled training data across the three detector families, as an augmentation of small real datasets, and as a label-free

threshold-calibration signal; finally, its controllability is exploited to map detector performance across fault severity and duration. Phases 1 and 2 are reported in Chapter 3, Phase 3 in Chapter 4, and Phase 4 in Chapter 5.

2.7. Motivating Simulation: the Drift-Fault Detection Gap

To ground the detection problem in a concrete spacecraft-engineering setting, before turning to the data-driven detectors, a closed-loop attitude-dynamics simulation was built in MATLAB Simulink. Its purpose is to reproduce, under fully controlled conditions, the structural failure of classical variance-based FDIR on a drift fault, the fault family that corresponds to the contextual anomaly class of the dataset.

The model implements the rigid-body rotational dynamics of a spacecraft,

$$\mathbf{I}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}) = \boldsymbol{\tau}, \quad (2.4)$$

where $\boldsymbol{\omega}$ is the angular-velocity vector, $\mathbf{I}$ the inertia tensor and $\boldsymbol{\tau}$ the applied control torque, with the inertia values taken from Crotti and Colagrossi [4]: $I_{xx} = 1.000$, $I_{yy} = 0.250$, $I_{zz} = 0.950$ kg m². A proportional–derivative detumbling controller drives the angular rate toward zero, a gyroscope sensor model adds band-limited measurement noise, and a fault-injection block introduces a drift fault on the gyroscope x -axis at $t = 300$ s. A classical FDIR variance monitor, following the running-variance approach of Colagrossi and Lavagna [3], observes the gyroscope signal and is intended to raise a flag when the running variance exceeds a calibrated threshold.

During the nominal phase the controller successfully damps the initial tumbling rate of approximately 10 deg/s toward zero, confirming correct closed-loop behaviour. After the drift fault is injected at $t = 300$ s, the sensor estimate diverges progressively from the true rate, the controller, acting on corrupted measurements, applies increasingly incorrect torques, and the estimation error grows continuously to about 14 deg/s by the end of the simulation. The critical result is in the variance panel of Figure 2.2: the running variance of the gyroscope signal stays below the detection threshold throughout the entire fault period, and the FDIR flag is never raised. This is structural, not an artefact of the particular threshold chosen. A slow, monotonic drift adds an almost constant offset within any short sliding window, so the running variance it produces stays near its nominal level wherever the threshold is set; only a threshold low enough to trip on nominal noise would catch it. The drift accumulates too slowly to leave any variance signature, and so

escapes classical detection completely, even as it drives a severe and growing degradation of attitude control.

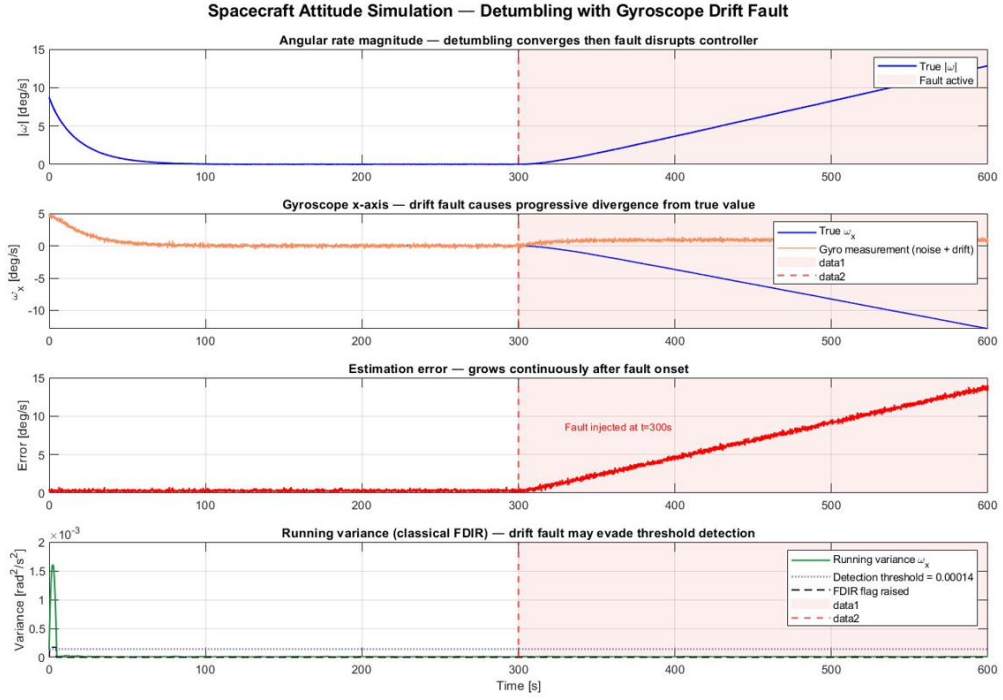


Figure 2.2: Closed-loop attitude-dynamics simulation of detumbling with a gyroscope drift fault injected at $t = 300$ s. The controller detumbles the spacecraft correctly during nominal operation; after fault onset the gyroscope estimate diverges from the true angular rate and the estimation error grows to approximately 14 deg/s. The running variance used by the classical FDIR monitor (following Colagrossi and Lavagna [3]) remains below its detection threshold throughout the fault period, so the fault is never flagged. Inertia values from Crotti and Colagrossi [4]; the drift rate and detection threshold are illustrative values chosen to expose the detection gap.

This simulation gives direct, controlled evidence for the gap identified in the literature of Chapter 1: a slowly accumulating fault produces no variance signature a static threshold can catch, yet causes a serious and growing control error. The detector developed in Chapter 3 must instead respond to the change in the temporal pattern of the signal, the distributional or predictive deviation the drift induces well before any amplitude bound is breached. With the problem framed in spacecraft-engineering terms and the evaluation protocol fixed, the next chapter establishes the detection baselines and the data-driven detectors on real telemetry.

3 | Detection Baselines on Real Spacecraft Telemetry

This chapter reports the first two phases of the methodology. Section 3.1 establishes the classical detection floor with a z-score threshold detector (Phase 1). Section 3.2, Section 3.3 and Section 3.4 build the three data-driven detectors of Phase 2: a supervised classifier, an unsupervised Variational Autoencoder, and an LSTM regression network. Section 3.5 then draws the four methods together in a unified comparison, with the SMAP channel P-1 case study, and Section 3.6 interprets the results and frames the question that Phase 3 addresses. Every quantity reported in this chapter was computed directly from the experimental result files under the point-adjust protocol of Section 2.3.

3.1. Phase 1 — The Classical Threshold Baseline

The baseline detector is the simplest form of statistical threshold detection, and is conceptually equivalent to the running-variance approach of classical FDIR [3]. A test time step is flagged as anomalous when its value deviates from the nominal training mean by more than three standard deviations,

$$a_t = \mathbb{1}[|x_t - \mu_{\text{train}}| > 3\sigma_{\text{train}}], \quad (3.1)$$

with μ_{train} and σ_{train} estimated on the nominal training split. The detector serves two purposes: it fixes the quantitative floor that the data-driven methods must exceed, and it confirms empirically that amplitude-based detection is structurally inadequate for the distributional anomalies that dominate real spacecraft telemetry.

Across the 81 unique channels the threshold detector achieves a mean F_1 of 0.183 and a median of only 0.019. The median is the more informative figure: the typical channel scores under two per cent, meaning the detector either never triggers or triggers so rarely that it misses the overwhelming majority of real fault events. Of the 81 channels, 35 score an F_1 of exactly zero, the threshold never activating despite verified anomalies in the

ground truth, and a further 15 score below 0.05. On the 30 contextual-anomaly channels the mean F_1 collapses to 0.052, against 0.259 on the 51 point-anomaly channels. This is exactly the asymmetry the fault taxonomy predicts: amplitude thresholds catch some point anomalies but are blind to contextual ones.

The channels on which the baseline performs well are revealing. The four strongest, D-4 ($F_1 = 0.998$), D-7 (0.981), D-12 (0.969) and D-3 (0.965), all have anomaly rates above 34% and anomaly windows that are large, sustained amplitude deviations reliably crossing the three-sigma boundary. These are structurally simple cases, where the fault is amplitude-separable from the nominal distribution. They are not representative of the detection challenge. The 35 zero-scoring channels, and the contextual channels generally, are the cases where the fault stays within normal amplitude bounds while the statistical pattern shifts. Channel P-1, the primary illustrative case, scores $F_1 = 0.011$, consistent with Figure 2.1, in which all three anomaly windows are contextual and produce no amplitude exceedance. These results fix a concrete floor: any improvement on the mean of 0.183 and the median of 0.019, achieved without labelled fault data, is a measurable advance beyond classical detection.

3.2. Phase 2, Step A — Supervised Telemetry Classification

Step A treats anomaly detection as supervised binary classification. Each 64-step window is labelled nominal or anomalous according to its overlap with the ground-truth windows (Section 2.4), and a classifier is trained to predict the label from six hand-crafted time-domain features: the window mean, standard deviation, minimum, maximum, range, and the lag-one autocorrelation. The first five features summarise the amplitude distribution; the lag-one autocorrelation is the feature that matters most for contextual anomalies, because a fault that alters the oscillation rhythm of a signal changes its autocorrelation even when its amplitude statistics are unchanged. This is the first application of the SVM and Multilayer Perceptron architectures benchmarked on simulator data by Crotti and Colagrossi [4] to real in-orbit telemetry. The Support Vector Machine uses a radial-basis-function kernel with $C = 10$ and balanced class weights; the Artificial Neural Network is a two-hidden-layer MLP (32 and 16 units), matching the depth evaluated by Crotti and Colagrossi [4]. A leave-one-out cross-validation protocol over the anomalous windows of each channel guarantees an evaluable, class-balanced fold for every channel regardless of how few anomalies it contains; full hyperparameters are listed in Appendix A.

The SVM achieves a mean F_1 of 0.598 and a median of 0.690, with non-zero detection on every one of the 81 channels, an improvement of +0.415 in mean and +0.671 in median over the threshold baseline. The MLP is weaker, at mean $F_1 = 0.413$ and median 0.253. Disaggregated by fault class, the SVM reaches 0.508 on contextual channels, nearly a tenfold gain over the threshold baseline’s 0.052, and 0.651 on point channels; the MLP reaches 0.344 and 0.453 respectively. Figure 3.1 shows the per-channel distribution for both classifiers. The contextual-channel result confirms the design hypothesis: the lag-one autocorrelation feature gives the SVM a discriminative signal on the temporal-pattern change that amplitude-only detection lacks entirely.

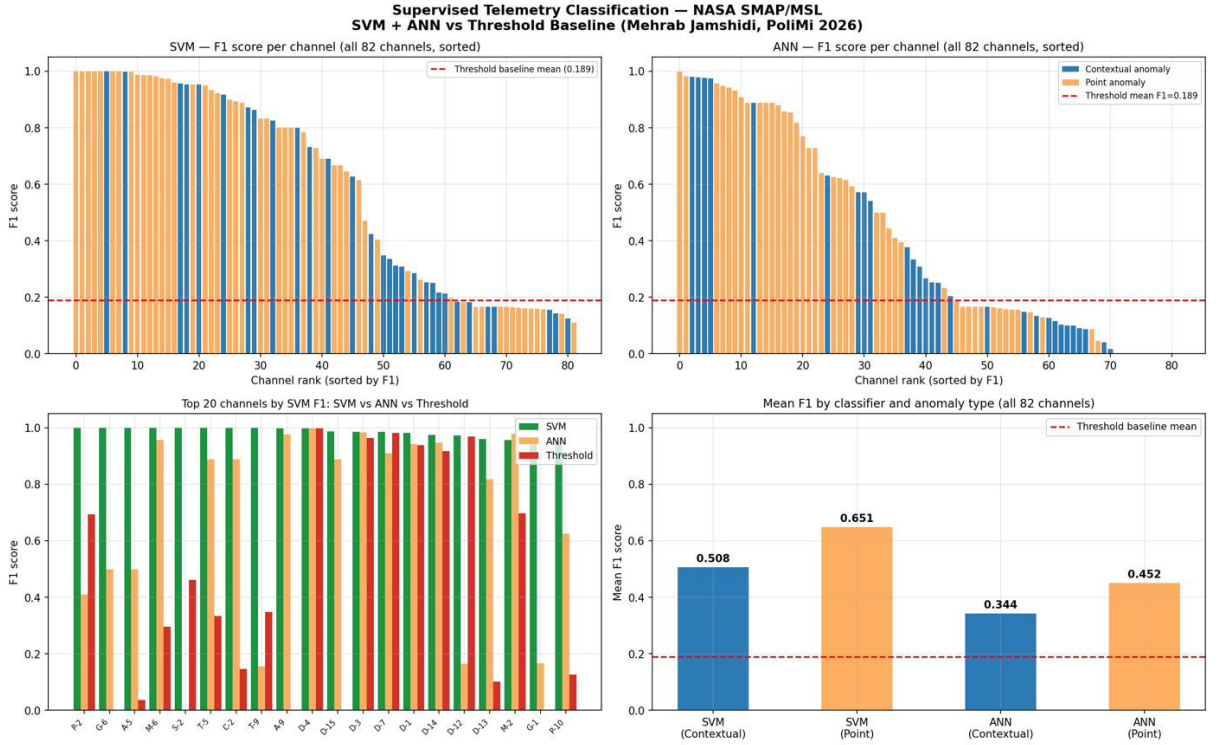


Figure 3.1: Supervised classification across the 81 unique channels. Per-channel F_1 for the SVM and the MLP, with contextual channels distinguished from point channels and the threshold-baseline mean shown for reference. The SVM achieves non-zero detection on every channel and a mean F_1 of 0.598.

Step A depends, however, on labelled anomaly examples for training through the leave-one-out protocol. In a real FDIR deployment before any in-orbit fault has occurred, such labels do not exist. This limitation motivates the unsupervised detectors of Section 3.3 and Section 3.4, which require no labelled faults, and ultimately the generative augmentation of Chapter 4, which synthesises labelled examples when real ones are unavailable.

3.3. Phase 2, Step B — Variational Autoencoder

Step B performs unsupervised detection with a Variational Autoencoder trained only on the nominal training split of each channel; no labelled fault data is used at any stage of training. The encoder compresses a 64-step window to an eight-dimensional latent distribution parameterised by a mean μ and log-variance $\log \sigma^2$, and the decoder reconstructs the window from a latent sample. Training through the stochastic sampling step uses the reparameterisation trick, $z = \mu + \epsilon \odot \exp(\frac{1}{2} \log \sigma^2)$ with $\epsilon \sim \mathcal{N}(0, I)$, and the objective is the standard evidence lower bound,

$$\mathcal{L}_{\text{VAE}} = \|x - \hat{x}\|^2 + \beta \text{KL}[\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, I)], \quad \beta = 1, \quad (3.2)$$

in which the reconstruction term drives faithful reproduction of nominal windows and the Kullback–Leibler term keeps the latent distribution close to a standard-normal prior, yielding a smooth, organised latent space. At inference the window is encoded deterministically to its latent mean and reconstructed, and the per-window reconstruction error is the anomaly score, max-pooled to per-time-step scores and evaluated under the point-adjust protocol.

Under point-adjust the VAE achieves a mean F_1 of 0.590 at the oracle threshold and 0.417 at the deployable $k = 3$ threshold, with non-zero detection on 76 and 58 channels respectively, against 46 for the threshold baseline. The most significant result for the thesis argument is the contextual-channel performance: the VAE reaches a mean F_1 of 0.638 on the contextual channels at the oracle threshold, more than twelve times the threshold baseline’s 0.052 under the point-adjust protocol, and trained entirely without labelled fault data. The mechanism is direct. During training the encoder learns to map the normal oscillation pattern of a channel into a compact region of latent space. A contextual anomaly, whose temporal structure departs from nominal despite staying within normal amplitude bounds, maps to an unusual latent code that the decoder, trained only on normal patterns, cannot reconstruct accurately, and the elevated reconstruction error is the detection signal. The VAE is in fact stronger on contextual channels (0.638) than on point channels (0.561), which confirms that its distributional learning is specifically suited to the pattern-shift faults that define the contextual class. Figure 3.2 shows the per-channel F_1 distribution across all 81 channels.

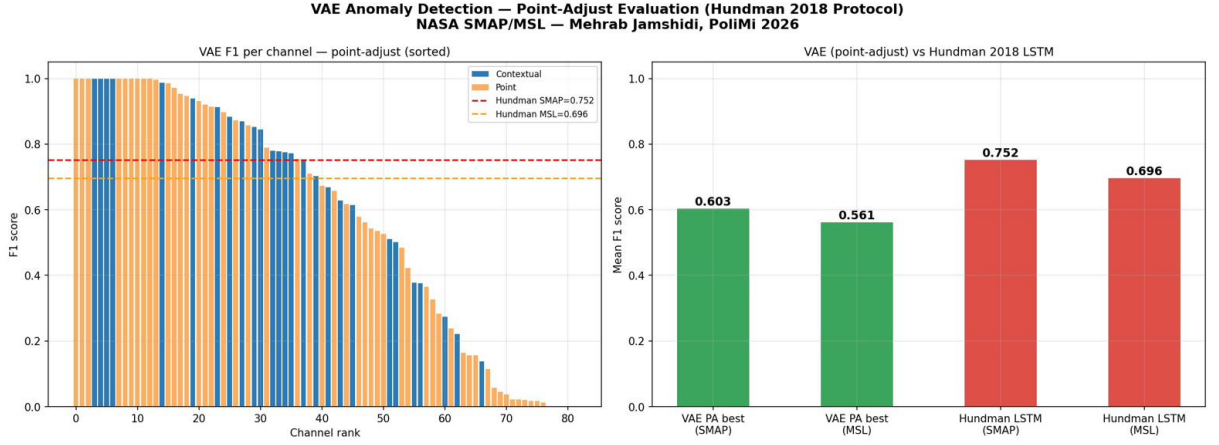


Figure 3.2: VAE anomaly detection under point-adjust across the 81 channels, with the published LSTM benchmark of Hundman et al. [15] shown for reference (SMAP 0.752, MSL 0.696). The VAE achieves a mean F_1 of 0.590 at the oracle threshold, with particular strength on contextual channels.

On channel P-1 the VAE reaches $F_1 = 0.275$ at the oracle threshold but 0.000 at $k = 3$. Figure 3.3 shows why: the reconstruction error is elevated inside the three anomaly windows, but only moderately above the training noise floor, so the separation is real yet insufficient for the conservative three-sigma threshold. This subtlety motivates the predictive approach of Section 3.4, which models the causal direction of time and is therefore more sensitive to the change in dynamics that defines P-1’s anomalies.

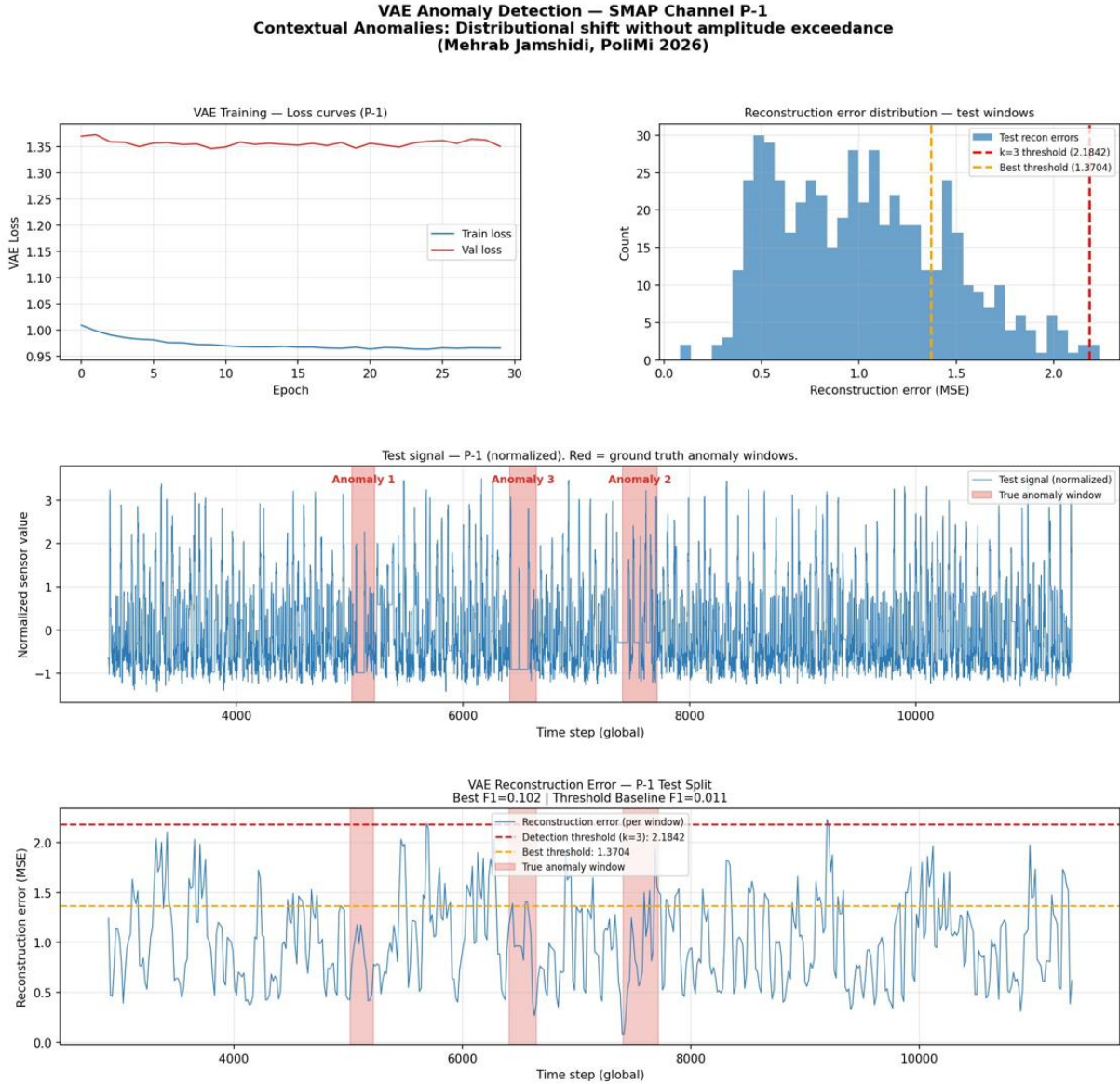


Figure 3.3: VAE detail for channel P-1. The reconstruction error is elevated within the three contextual anomaly windows but only moderately above the nominal training-error distribution, giving partial detection at the oracle threshold ($F_1 = 0.275$) and none at $k = 3$.

3.4. Phase 2, Step C — LSTM Regression

Step C performs detection by telemetry regression: a two-layer LSTM network [14] with 64 hidden units per layer is trained to predict the next sensor value from the preceding

64 time steps, and the one-step-ahead squared prediction error,

$$e_t = (\hat{x}_t - x_t)^2, \quad \hat{x}_t = f_{\text{LSTM}}(x_{t-64}, \dots, x_{t-1}), \quad (3.3)$$

is the anomaly score. The essential distinction from the VAE is temporal directionality. The VAE compresses and reconstructs a window without regard to the causal order of its values; the LSTM processes the window sequentially, accumulating a hidden state that summarises the signal’s dynamics up to each step, so its prediction is conditioned on the entire preceding history. When a fault alters the temporal dynamics of the signal, as contextual anomalies do, the prediction is wrong, and the error measures the dynamic change directly. By adopting the same model family as Hundman et al. [15], this step also provides an independent reproduction of the published benchmark that validates the entire experimental pipeline.

The LSTM is the strongest detector evaluated, with a mean F_1 of 0.743 at the oracle threshold and 0.661 at $k = 3$, and a median of 0.918 at the oracle threshold; more than half the channels exceed $F_1 = 0.90$, a level that, as noted in Section 2.3.2, reflects the permissiveness of point-adjust on long segments as much as raw detector skill. The comparison with Hundman et al. [15] is close and validating. On the 54 unique SMAP channels (the 55 SMAP channels of Section 2.2.1 less the duplicated P-2) the LSTM reaches a mean F_1 of 0.746 against the published 0.752, a gap of only 0.006, and on the 27 MSL channels it reaches 0.738, exceeding the published 0.696 by 0.042. The residual SMAP gap is a matter of threshold calibration: Hundman et al. [15] use a nonparametric dynamic thresholding procedure, whereas this implementation uses the simpler k -sigma rule. That the optimal calibration is highly channel-specific is clear from the distribution of best thresholds: 27 channels are best served by the 99th percentile of training error, 24 by $k = 1.0$, and 21 by $k = 4.0$, with the rest scattered between. That spread is what motivates the threshold-calibration study of Chapter 5.

On contextual channels the LSTM reaches a mean F_1 of 0.836, the strongest method-class combination in the study, exceeding both Hundman benchmark values and the VAE’s contextual F_1 of 0.638 by 0.198. This confirms the structural advantage of causal temporal modelling for faults whose signature is a change in dynamics rather than amplitude. Figure 3.4 shows the per-channel distribution, and Figure 3.5 the detail for P-1: the one-step-ahead prediction tracks the signal closely during nominal operation and visibly diverges inside each anomaly window, lifting the prediction error above the $k = 3$ threshold and yielding $F_1 = 0.726$ deployable and 0.814 at the oracle threshold, against the baseline’s 0.011.

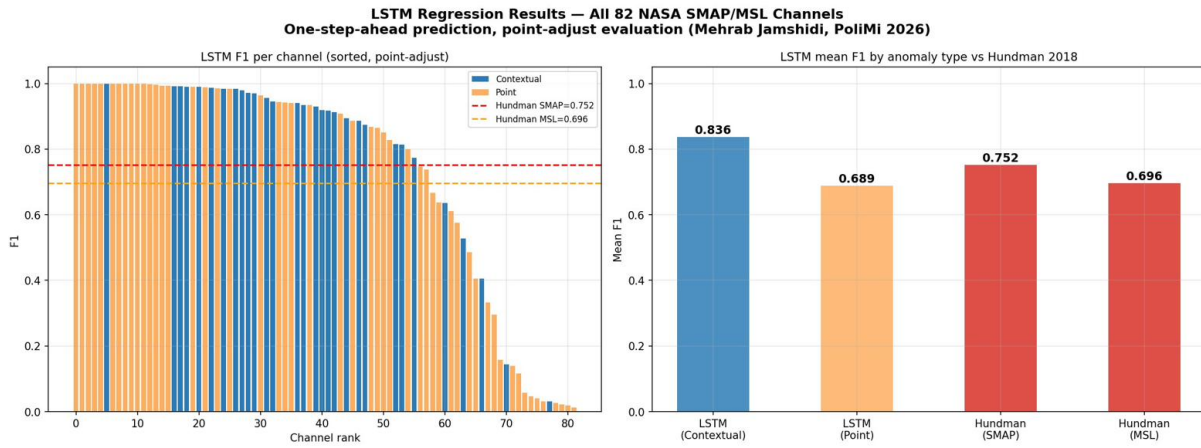


Figure 3.4: LSTM regression under point-adjust across the 81 channels. A majority of channels exceed both Hundman benchmark reference lines; the LSTM contextual-anomaly mean F_1 of 0.836 substantially exceeds both published values.

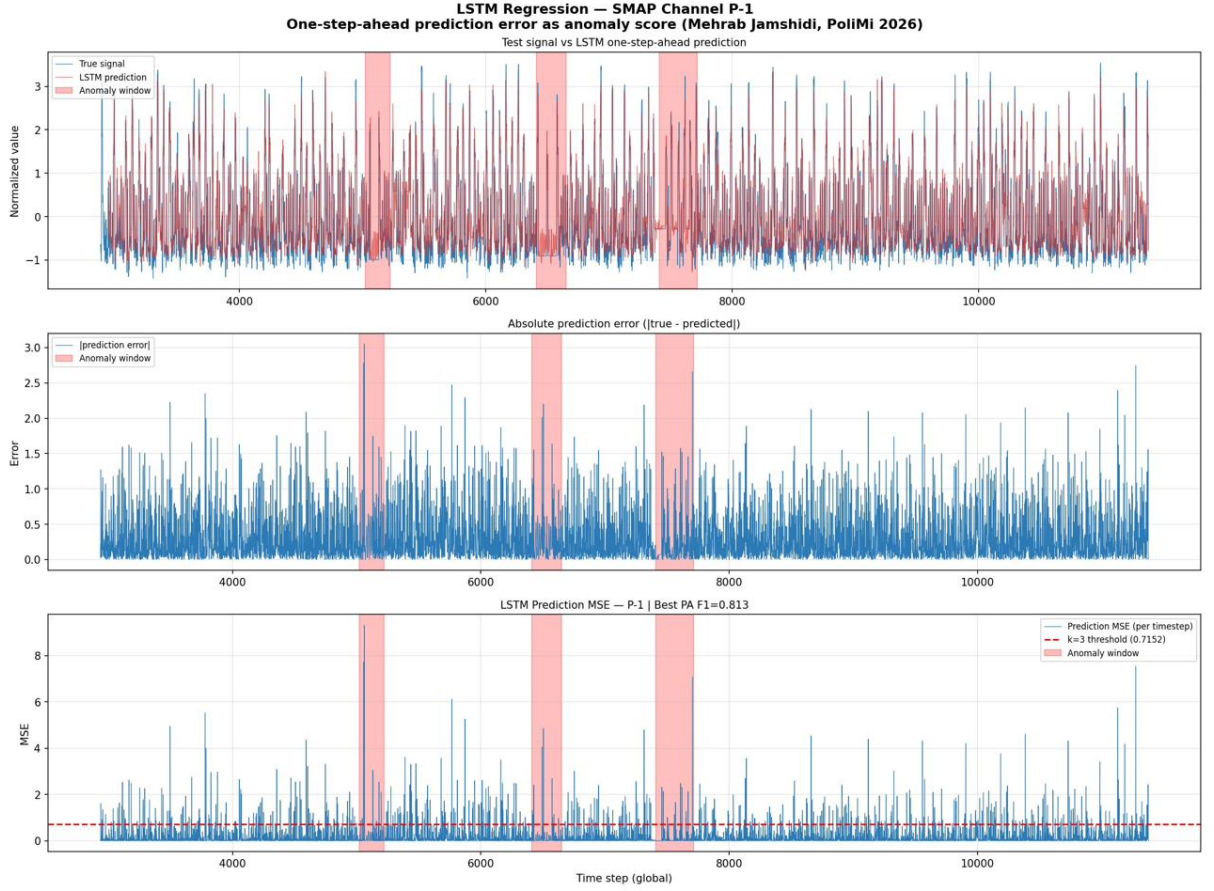


Figure 3.5: LSTM detail for channel P-1. The one-step-ahead prediction tracks the true signal during nominal operation and diverges inside all three contextual anomaly windows; the prediction error exceeds the $k = 3$ threshold, giving $F_1 = 0.726$ deployable and 0.814 at the oracle threshold, against the threshold baseline’s 0.011.

3.5. Complete Method Comparison

Table 3.1 presents the four methods together under the unified point-adjust protocol, with the published values of Hundman et al. [15] for reference. The LSTM at the oracle threshold is the strongest overall, but the most operationally important comparison is between the SVM and the VAE: they reach almost identical aggregate F_1 at the oracle threshold (0.598 and 0.590); the VAE needs no labelled fault data to train — unlike the SVM’s leave-one-out supervision — and is stronger on the contextual class. At the deployable $k = 3$ threshold its aggregate F_1 falls to 0.417, so this advantage is about label-free training and contextual strength, not aggregate parity.

Method	Mean F_1	Median F_1	$F_1 > 0$	$F_1 > 0.5$
Z-score threshold (baseline)	0.183	0.019	46 / 81	12 / 81
ANN classifier (MLP 32 $\rightarrow$ 16)	0.413	0.253	70 / 81	32 / 81
VAE — point-adjust ($k = 3$)	0.417	0.327	58 / 81	37 / 81
SVM classifier (RBF kernel)	0.598	0.690	81 / 81	46 / 81
VAE — point-adjust (oracle)	0.590	0.674	76 / 81	52 / 81
LSTM regression ($k = 3$)	0.661	0.849	78 / 81	55 / 81
LSTM regression (oracle)	0.743	0.918	81 / 81	63 / 81
Hundman et al. (2018) LSTM — SMAP [†]	0.752	—	—	—
Hundman et al. (2018) LSTM — MSL [†]	0.696	—	—	—

Table 3.1: Detection performance across all 81 unique channels under point-adjust. [†] Published values from Hundman et al. [15], included for reference.

Table 3.2 disaggregates the results by fault class and spacecraft. The contextual column carries the central message: the threshold baseline manages only 0.052, while the LSTM reaches 0.836, a sixteenfold improvement, under point-adjust, on the fault class that classical FDIR cannot detect, and the VAE reaches 0.638 — both at the oracle threshold, and trained with no labelled fault data. The complete per-channel F_1 for every detector is reported in Chapter B.

Method	Contextual ($n=30$)	Point ($n=51$)	SMAP ($n=54$)	MSL ($n=27$)
Threshold	0.052	0.259	0.169	0.210
SVM	0.508	0.651	0.534	0.726
ANN	0.344	0.453	0.435	0.369
VAE (oracle)	0.638	0.561	0.604	0.561
LSTM (oracle)	0.836	0.688	0.746	0.738

Table 3.2: Mean F_1 disaggregated by anomaly class and spacecraft, under point-adjust.

Finally, Table 3.3 collects the per-method results for channel P-1, whose three contextual anomaly windows are structurally invisible to amplitude thresholding. The progression

from $F_1 = 0.011$ (threshold) to 0.814 (LSTM) on this channel encapsulates the central empirical argument of the thesis.

Method	Precision	Recall	F_1
Z-score threshold	0.040	0.007	0.011
SVM classifier	0.105	0.702	0.182
ANN classifier	0.076	0.246	0.116
VAE (oracle threshold)	0.206	0.412	0.275
LSTM ($k = 3$ threshold)	0.570	1.000	0.726
LSTM (oracle threshold)	0.686	1.000	0.814

Table 3.3: Channel P-1 results across all methods. P-1 contains three contextual anomaly windows with no amplitude exceedance; the test-split anomaly rate is 8.8%.

3.6. Discussion

Three conclusions emerge. First, supervised classification with temporal features substantially outperforms classical amplitude detection across all channels and fault classes, confirming on real spacecraft telemetry the conclusion reached by Crotti and Colagrossi [4] on simulator data, with the lag-one autocorrelation feature as the enabling factor for contextual detection. Second, the unsupervised VAE matches the supervised SVM in aggregate at the oracle threshold (0.590 against 0.598) and surpasses it on the contextual class, while needing no labelled fault data to train; at the deployable $k = 3$ threshold its aggregate F_1 is 0.417, below the SVM, so the VAE’s advantage is its label-free training and contextual strength rather than aggregate parity. Third, the LSTM is the strongest detector and, through its agreement with the published benchmark of Hundman et al. [15] to within 0.006 on SMAP, validates the entire experimental pipeline.

The contrast between the VAE and the LSTM on contextual channels (0.638 against 0.836) reflects a meaningful architectural distinction. The VAE models the distributional structure of windows, what a normal window looks like, while the LSTM models the causal dynamics, how the signal evolves. Contextual anomalies are fundamentally violations of causal dynamics: the signal keeps looking superficially normal while its rhythm or correlation structure changes, and the LSTM’s sequential, direction-sensitive processing is structurally matched to that. Both, however, remain below the state-of-the-art Anomaly Transformer of Xu et al. [32], which reports F_1 above 0.90 on both spacecraft; that gap

defines the upper-bound reference for the remainder of the thesis.

Mapped onto the Digital Twin maturity framework of Wei et al. [31], these results trace a clear progression. The threshold detector corresponds to Level 3 (Operational): automated monitoring with no learning. The supervised SVM and MLP, trained on real labelled telemetry, represent Level 4 (Prescriptive): data-driven adaptation of the decision boundary. The VAE and LSTM, which learn autonomously from nominal data and could be updated continuously as new nominal telemetry arrives, represent the learning component of Level 5 (Cognitive). In terms of the fault taxonomy of Colagrossi and Lavagna [3], the drift and erratic faults that escape classical FDIR, demonstrated in the simulation of Section 2.7 and corresponding to the contextual class, are now detected by all three machine-learning methods, with the LSTM reaching 0.836 on the contextual channels. The detection gap identified in the simulation is quantitatively closed on real spacecraft telemetry.

Three limitations qualify these results. First, only the primary sensor measurement (column 0) is used; incorporating correlated channels would likely improve detection of contextual anomalies whose signature spans multiple sensors, and is the highest-priority architectural extension. Second, the oracle-threshold figures require test labels and are upper bounds; the $k = 3$ figures are the fair, deployable comparison, and the gap between them, 0.173 for the VAE and 0.082 for the LSTM, is exactly the deployable-calibration headroom that Chapter 5 examines. Third, the leave-one-out SVM evaluation produces small per-fold sets on channels with very few anomalies, so the occasional $F_1 = 1.0$ result should be read as indicative rather than definitive.

These detectors share one decisive operational dependence: the supervised SVM requires labelled fault examples that an operational spacecraft may not possess before its first in-orbit anomaly. The next chapter confronts that dependence directly, training a generative model to synthesise the fault data that real deployment lacks.

4 | Conditional Generative Modelling of Fault Data

The detectors of Chapter 3 share one operational dependence: the supervised classifier requires labelled fault examples that an operational spacecraft may not possess before its first in-orbit anomaly, and even the unsupervised detectors are calibrated against an assumed fault distribution. This chapter introduces the first generative component of the thesis (Phase 3): a conditional Wasserstein Generative Adversarial Network with Gradient Penalty, trained on the real anomaly windows of the 61 training channels held aside under the clean protocol of Section 2.5, whose purpose is to synthesise realistic fault data that can stand in for the labelled telemetry that real deployment lacks. Because the 20 held-out channels are excluded from the generator’s training pool, the train-on-synthetic transfer result of Section 4.6 is computed on anomalies the generator has never seen. The distributional realism metrics of Section 4.5, by contrast, compare the synthetic windows against the real anomaly windows on which the generator was trained: they measure how faithfully the generator reproduces its own training distribution and are therefore an in-sample fidelity check, not a held-out test. The held-out evaluation in this chapter is the transfer result alone. Section 4.1 reviews the theoretical background; Section 4.2 sets out the four architectural contributions that distinguish this model from a standard WGAN-GP; Section 4.3 specifies the architecture and training protocol; and Section 4.5 and Section 4.6 evaluate the realism of the synthetic output distributionally and through a downstream detection task. As in the preceding chapter, every reported quantity was computed directly from the experimental result files.

4.1. Theoretical Background

4.1.1. Generative adversarial networks and their failure modes

A Generative Adversarial Network [10] consists of a generator G , which maps a noise vector $z \sim \mathcal{N}(0, I)$ to a synthetic sample, and a discriminator, which attempts to distinguish

synthetic samples from real ones. The two are trained in opposition until the generator’s output distribution is indistinguishable from the data distribution. In its original form this two-player game is notoriously unstable. The two failure modes most damaging in the small-data regime of this thesis are vanishing gradients and mode collapse. In the first, the discriminator becomes too strong and provides no useful gradient to the generator; in the second, the generator maps many noise vectors to a few outputs, capturing only part of the data distribution. With only a few dozen real anomaly windows per class, both are acute, and a more robust training formulation is required.

4.1.2. Wasserstein GAN with gradient penalty

The Wasserstein GAN [1] replaces the discriminator with a *critic* that estimates the Wasserstein (earth-mover’s) distance between the real and generated distributions. Unlike the original objective, the Wasserstein distance provides a smooth, meaningful gradient even when the two distributions do not overlap, directly addressing the vanishing-gradient problem. The estimate is valid only if the critic is 1-Lipschitz, and the gradient-penalty formulation of Gulrajani et al. [12] enforces this by penalising departures of the critic’s gradient norm from unity at points interpolated between real and generated samples. The critic objective is

$$\mathcal{L}_{\text{critic}} = \mathbb{E}_z[D(G(z))] - \mathbb{E}_x[D(x)] + \lambda_{\text{GP}} \mathbb{E}_{\hat{x}}[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2], \quad (4.1)$$

with $\hat{x} = \alpha x + (1 - \alpha)G(z)$, $\alpha \sim U[0, 1]$, and $\lambda_{\text{GP}} = 10$. The critic is updated $n_{\text{critic}} = 5$ times per generator update, the standard ratio of Gulrajani et al. [12], so that the Wasserstein estimate remains accurate as the generator improves. The WGAN-GP formulation has become the dominant variant for fault-data synthesis precisely because it is stable under the severe class imbalance characteristic of the domain [18].

4.1.3. Conditional generation and stabilisation

Because the dataset contains two structurally distinct fault classes, the generator must be conditioned on the class label so that point and contextual windows can be synthesised separately. Two further techniques stabilise training. Spectral normalization [24] constrains the spectral norm of each critic weight matrix to one, enforcing the Lipschitz constraint by construction and complementing the gradient penalty. The two time-scale update rule [13] gives the critic a higher learning rate than the generator, letting the critic track the moving data distribution while the generator changes more slowly, which improves convergence in practice. Both are adopted here.

4.2. Architectural Contributions

This chapter contributes four architectural choices beyond a standard conditional WGAN-GP, each motivated by a specific failure of the standard formulation on this dataset.

One-dimensional convolutional architecture. An early multilayer-perceptron implementation, which processes each time step independently through fully connected layers, destroys the local temporal structure that distinguishes a coherent telemetry signal from noise. The generator and critic here are instead fully convolutional in one dimension: the generator upsamples a noise vector to a length-64 window through transposed convolutions, and the critic downsamples a window through strided convolutions. The convolutional kernels share weights across time and therefore learn local temporal patterns directly, which is essential for reproducing the oscillation structure that defines the contextual fault class.

Autocorrelation-matching loss. The adversarial objective alone does not guarantee that the temporal correlation structure of the synthetic windows matches the real windows. An explicit autocorrelation-matching term is therefore added to the generator loss, penalising the squared difference between the autocorrelation functions of generated and real windows over lags 1 to 20:

$$\mathcal{L}_{\text{gen}} = -\mathbb{E}_z[D(G(z))] + \lambda_{\text{ACF}} \|\text{ACF}(G(z)) - \text{ACF}_{\text{real}}\|^2, \quad \lambda_{\text{ACF}} = 5, \quad (4.2)$$

where ACF_{real} is computed once from the real windows of the relevant class and held fixed. The autocorrelation is implemented differentiably so that the term contributes a gradient to the generator. This provides direct supervision of temporal structure, in addition to the indirect pressure exerted by the critic, and is the most distinctive of the four contributions in that it acts on the temporal-correlation structure explicitly rather than leaving it to emerge from the adversarial objective. Its marginal effect was not isolated in a controlled ablation against the three other contributions; quantifying each contribution separately is identified as a refinement in Section 4.7.

Per-class specialised generators. A single conditional generator covering both fault classes on so small a dataset tends to average across the two structurally distinct distributions, producing windows characteristic of neither. The model instead trains two specialised generators, one per class, eliminating mode-averaging at the cost of a modest increase in total parameters. This makes class conditioning explicit at the architectural

level rather than relying on a label input that the generator may learn to ignore.

Spectral normalization over batch normalization in the critic. Batch normalization in the critic is mathematically incompatible with the gradient-penalty term, because the penalty requires the gradient of the critic with respect to each individual interpolated sample, whereas batch normalization couples samples within a batch. All learnable critic layers therefore use spectral normalization instead, providing immediate Lipschitz-constraint enforcement that complements rather than conflicts with the gradient penalty.

4.3. Model Architecture and Training Protocol

The generator is a four-layer one-dimensional convolutional network. A 100-dimensional noise vector is reshaped to a length-one sequence and expanded through three transposed-convolution layers (kernel size 8, stride 4), each followed by batch normalization and a LeakyReLU activation with negative slope 0.2, after which a final convolution with a tanh activation produces a length-64 univariate window in $[-1, +1]$; the generator contains 184,865 parameters. The critic is a four-layer convolutional network with spectral normalization on every learnable layer: three strided convolutions (kernel size 4, stride 2) with LeakyReLU activations reduce the temporal dimension and increase the channel count, and a final linear layer projects to the scalar Wasserstein score; the critic contains 42,337 parameters. No batch normalization is used in the critic, for the reason given above. The complete layer-by-layer specifications are listed in Appendix A.

Training uses the Adam optimiser with $\beta_1 = 0$ and $\beta_2 = 0.9$, the reduced first-moment coefficient suppressing the momentum oscillation characteristic of adversarial training, with a generator learning rate of 1×10^{-4} and a critic learning rate of 4×10^{-4} in accordance with the two time-scale update rule. After the 20 held-out channels are excluded, the point generator is trained on 2,000 real point-anomaly windows and the contextual generator on 792 real contextual-anomaly windows, all drawn from the 61 training channels. Each generator is trained for 3,000 generator steps (equivalently 15,000 critic updates at $n_{\text{critic}} = 5$) with a batch size of 64 and the fixed random seed 42 used throughout the thesis. Table 4.1 summarises the configuration; the full justification of each value follows the protocol of Gulrajani et al. [12] and Heusel et al. [13].

Hyperparameter	Value
Latent dimension	100
Generator steps	3,000
Critic steps per generator step (n_{critic})	5
Gradient-penalty weight λ_{GP}	10
ACF-loss weight λ_{ACF}	5.0
Generator / critic learning rate	$1 \times 10^{-4} / 4 \times 10^{-4}$
Adam (β_1, β_2)	(0.0, 0.9)
Batch size	64
Synthetic windows generated per class	2,000
Random seed	42

Table 4.1: Training configuration of the conditional WGAN-GP. Two generators are trained, one per anomaly class, under identical settings.

4.4. Training Convergence

Training is stable for both classes from the first step. The gradient-penalty value falls rapidly from initial values near 7 to below 0.05 by step 200, the combined effect of spectral normalization, which enforces the Lipschitz constraint immediately, and the gradient penalty, which provides additional regularisation. The autocorrelation-matching loss falls from initial values near 0.12 to below 0.003 within the first 100 steps and remains near zero thereafter. This shows that the optimisation succeeds in minimising the term; it is, however, partly expected by construction, since the term is itself the squared autocorrelation mismatch that the optimiser directly minimises, so its decay is corroborating rather than independent evidence. The more telling observation is that this is achieved while the gradient penalty and adversarial losses also remain controlled, indicating that the convolutional architecture is able to reproduce temporally coherent output without the autocorrelation term and the adversarial objective working against one another. The two classes show characteristically different generator-loss dynamics: the contextual generator converges smoothly and stabilises by step 2,500, whereas the point generator reaches its minimum near step 800 and then drifts as the critic gradually identifies subtle distinguishing features of the generated point distribution. Throughout this drift the gradient penalty and ACF loss remain near zero, so the structural constraints continue to be sat-

ified even as the critic's relative confidence shifts. Figure 4.1 shows the training curves, and Figure 4.2 a selection of generated windows alongside real ones for both classes.

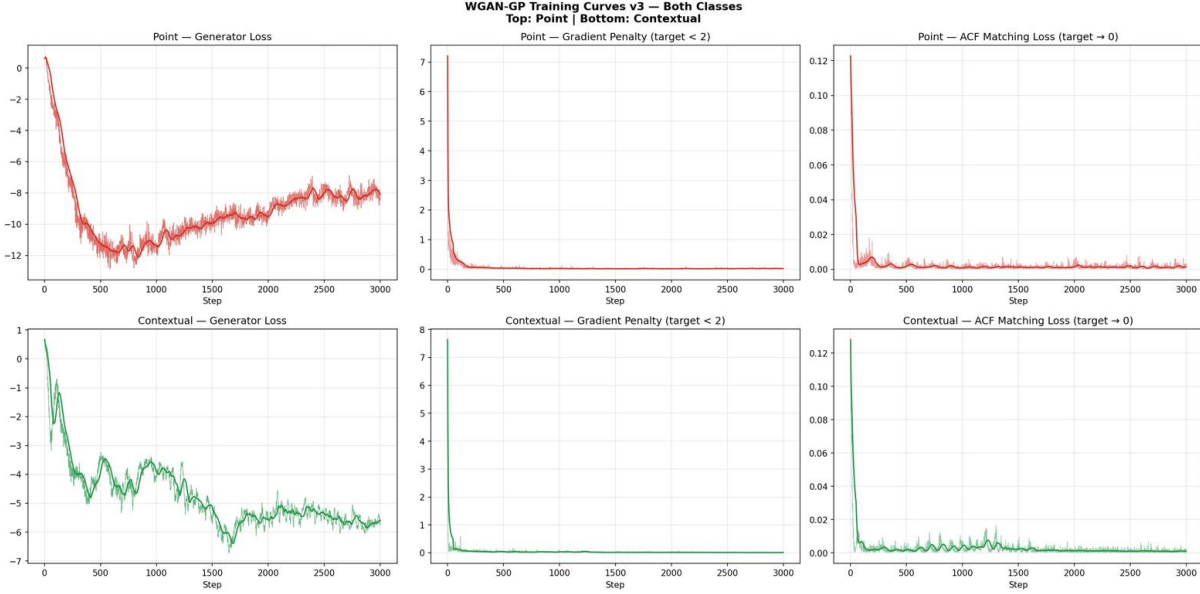


Figure 4.1: Training curves for both generators (top: point class; bottom: contextual class). The gradient penalty stabilises below 0.05 by step 200 and the autocorrelation-matching loss falls below 0.003 within 100 steps and remains near zero, while the generator and critic losses reflect the cleaner convergence of the contextual distribution.

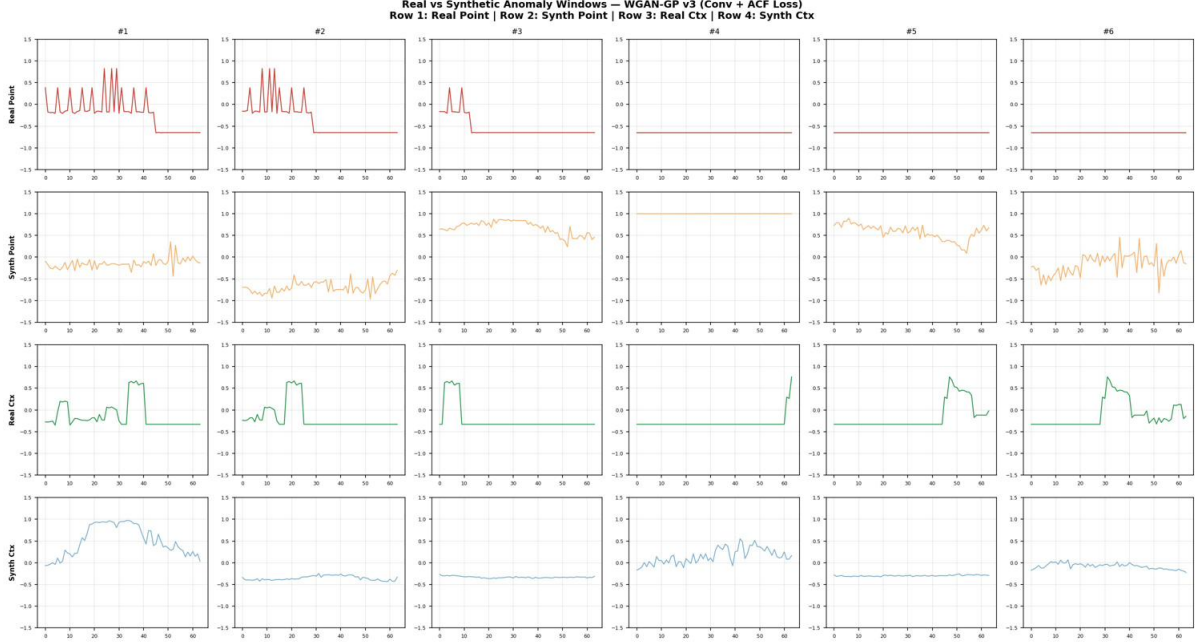


Figure 4.2: Real and synthetic anomaly windows for both classes. The synthetic contextual windows reproduce the oscillation structure of the real windows closely; the synthetic point windows capture the qualitative amplitude-excursion character with a residual gap on the most extreme excursions.

4.5. Generation Quality Evaluation

The realism of the synthetic windows is assessed with a suite of complementary distributional metrics, computed separately for each class against the real anomaly windows of that class that the generator was trained on. These are in-sample fidelity metrics: they quantify how closely the synthetic distribution matches the training distribution, and are distinct from the held-out transfer test of Section 4.6. The Fréchet Distance measures the distance between Gaussians fitted to the real and synthetic distributions in a six-feature evaluation space, by analogy with the Fréchet Inception Distance used for images. This six-feature space is the same hand-crafted feature representation used by the supervised classifier of Section 3.2, so the Fréchet Distance and the train-on-synthetic transfer result of Section 4.6 are related rather than fully independent checks: both reward a generator that matches the real windows in this feature space. They are reported together because they probe different consequences of that match, distributional overlap on the one hand and downstream decision-boundary transfer on the other, but neither should be read as independent confirmation of the other. The autocorrelation mean absolute deviation measures temporal-structure fidelity directly, as the mean absolute difference between the real

and synthetic autocorrelation functions over lags 1 to 20. These are complemented during development by a kernel Maximum Mean Discrepancy and by per-feature Kolmogorov–Smirnov statistics; for all of them, smaller values indicate closer agreement. Table 4.2 reports the two headline fidelity metrics, recomputed for the clean generator (the one trained only on the 61 training channels) so that they describe the generator that the transfer experiments of Chapter 5 actually use; the comparison is against that generator’s own real training windows, not the held-out anomalies.

Metric	Contextual	Point
Fréchet Distance (6-feature space)	0.141	0.403
ACF mean $ \Delta $ (lags 1–20)	0.026	0.015

Table 4.2: Generation-quality metrics by anomaly class for the clean generator (trained on the 61 training channels), computed against its real training windows; smaller is better. The contextual distribution is the more faithfully reproduced on the Fréchet metric, while both classes match the real autocorrelation structure closely.

Both classes are reproduced with small distributional error. The contextual class has the lower Fréchet Distance (0.141 against 0.403), and the autocorrelation deviation is small for both classes (0.026 for contextual and 0.015 for point), confirming that the explicit autocorrelation-matching term succeeds in transferring the real temporal-correlation structure to the synthetic windows. The contextual autocorrelation deviation is larger than in the development run that used all channels (it rose from 0.009 to 0.026), an expected consequence of moving several contextual channels into the held-out set, which shrank the contextual training pool; it remains small in absolute terms. A direct check of generation quality is the built-in train-on-synthetic preview, in which a classifier trained on the synthetic windows is scored on the real windows: excluding the held-out channels moved this preview only marginally, from 0.411 to 0.405, confirming that the held-out exclusion did not degrade the generator and that any change in the Phase 4 results can be attributed to the cleaner evaluation rather than to a weaker model.

The relative ordering of the two classes is instructive and partly reverses the detection difficulty established in Chapter 3. Contextual anomalies are subtle deviations that stay close to the nominal data manifold and share a consistent oscillatory structure, so the convolutional generator with its autocorrelation supervision reproduces their overall distribution precisely, giving the low Fréchet Distance. Point anomalies are large amplitude excursions whose form varies widely across the point channels of the training set, spikes,

steps and hardovers of differing magnitude, so a single per-class generator must cover a far more heterogeneous distribution; the higher Fréchet Distance reflects that heterogeneity rather than a training failure. Figure 4.3 shows a two-dimensional t-SNE projection consistent with this reading: the synthetic contextual samples overlap the real contextual samples closely, while the synthetic point samples occupy a partially overlapping but more concentrated region.



Figure 4.3: Two-dimensional t-SNE projection of real and synthetic windows. The synthetic contextual samples overlap the real contextual distribution closely; the synthetic point samples occupy a partially overlapping, more concentrated region, consistent with the heterogeneity of the real point distribution.

4.6. Train-on-Synthetic, Test-on-Real Validation

Distributional metrics establish that the synthetic windows resemble the real ones, but the operational question is whether they carry usable detection signal on faults the generator

has never seen. This is tested directly with the train-on-synthetic, test-on-real (TSTR) protocol under the clean held-out split: a Support Vector Machine, identical to that of Section 3.2, is trained exclusively on the synthetic windows and evaluated on the 20 held-out channels, whose anomalies never entered the generator’s training pool. A classifier that performs well under this protocol has learned, from synthetic data alone, decision boundaries that transfer to genuine, previously unseen faults.

The TSTR classifier achieves a mean F_1 of 0.474 on the 20 held-out channels, with a median of 0.452. The reference is the same classifier trained on real labelled data and evaluated on those same channels, which reaches 0.631; the synthetic-trained detector therefore retains about three quarters of the real-data performance, and a paired Wilcoxon signed-rank test across the 20 channels returns $p = 0.13$, so the difference is not statistically significant. On an unseen-channel test set, in other words, a detector trained purely on synthetic faults is not detectably worse than one trained on real faults; given the limited power of a 20-channel paired test, this is best read as the absence of a detectable difference rather than as equivalence. This is the central realism result of the generative phase, reported in full, with the per-detector substitution campaign and its caveats, in Chapter 5. Disaggregated by class, with the mixed channel C-1 grouped with the point class as throughout the thesis (Section 2.5.2), the held-out transfer reaches 0.57 on the 14 point channels and 0.24 on the 6 contextual channels. That gap follows the intrinsic detectability of each class on the real test set rather than the generation-quality ordering: the classifier’s amplitude-sensitive features handle the point class well, and point channels make up 14 of the 20 held-out set.

4.7. Discussion

Phase 3 demonstrates that a conditional WGAN-GP, trained on the real anomaly windows of the 61 training channels and equipped with the four architectural contributions of Section 4.2, produces synthetic spacecraft fault data of measurable quality and downstream utility. The headline results are that the contextual class is reproduced to a Fréchet Distance of 0.141 with both classes matching the real autocorrelation structure closely (0.026 contextual, 0.015 point), and that a classifier trained purely on synthetic data is not statistically distinguishable, on unseen held-out channels, from one trained on real data (0.474 against 0.631; Wilcoxon $p = 0.13$, $n = 20$). The autocorrelation-matching term is the most direct of the four contributions: it minimises the temporal-structure mismatch explicitly and is consistent with the faithful reproduction of the contextual oscillation structure, though, as noted in Section 4.4, its decay to near zero is partly expected by

construction and so does not on its own prove that the term is responsible for the quality of the output.

Four considerations bound the result. First, the model generates only the primary sensor measurement, inheriting the univariate restriction of the detectors in Chapter 3. Second, the per-class point generator must cover a heterogeneous mixture of amplitude-excursion sub-types across many channels, which accounts for the higher point-class Fréchet Distance and suggests that finer sub-type conditioning would help. Third, the four architectural contributions were adopted together and motivated qualitatively; their individual marginal effect was not isolated in a controlled ablation against a plain conditional WGAN-GP, and the realism evidence reported here is distributional and downstream rather than an exhaustive battery. In particular, no explicit nearest-neighbour test was run to rule out memorisation of training windows. A systematic ablation and a memorisation check are the natural next refinements of the generator evaluation. Fourth, the synthetic windows are validated here through a single supervised transfer task; whether they improve detection when used as a substitute for, or augmentation of, real training data across the full detector suite, and the conditions under which they help and those under which they do not, is the operational question this evaluation cannot answer on its own. That question is the subject of Chapter 5.

5 | Synthetic-to-Real Transfer and Sensitivity Analysis

Chapter 4 established that the conditional WGAN-GP produces synthetic fault data of measurable distributional quality, and that on unseen channels a classifier trained on synthetic faults is not significantly worse than one trained on real faults. This chapter (Phase 4) determines what that data is *worth* in operational terms. Every experiment is run under the clean held-out protocol of Section 2.5: the detectors are evaluated only on the 20 held-out channels, whose anomalies the generator never observed, and every synthetic-versus-real comparison is tested with a paired Wilcoxon signed-rank test over those 20 channels. Section 5.1 replaces real training data with synthetic data and measures the performance retained by each detector. Section 5.2 measures the marginal value of real fault examples and the effect of synthetic augmentation. Section 5.3 tests whether synthetic data can solve the deployable-threshold problem left open in Chapter 3, and resolves an apparent counter-example. Section 5.4 introduces the principal methodological contribution of the thesis: a sensitivity analysis that exploits the controllability of the generator to map detector capability across fault severity and duration. Section 5.5 consolidates the significance tests, and Section 5.6 synthesises the results. As before, every reported quantity was regenerated from the experimental result files.

5.1. Campaign 1 — Synthetic-to-Real Substitution

The first campaign tests the core hypothesis directly on unseen channels: each of the three Phase 2 detectors is retrained with the real labelled fault data replaced by synthetic anomaly windows from the Phase 3 generator, and evaluated on the 20 held-out channels. The like-for-like reference is the same detector trained on real data and evaluated on those same 20 channels, so the retained fraction measures how well the synthetic data substitutes for the real fault record on faults that were never seen during generation. Table 5.1 reports the result.

Detector	Real (held-out)	Synthetic (held-out)	Retention
SVM — supervised classifier	0.631	0.474	75%
VAE — unsupervised ($k = 3$)	0.397	0.397	100%
LSTM — unsupervised ($k = 3$)	0.547	0.543	99%

Table 5.1: Mean F_1 over the 20 held-out channels when real fault training data is replaced by synthetic data. The reference column is the same detector trained on real data and evaluated on the same channels. For the unsupervised detectors the deployable threshold is the classical $k = 3$ rule, which uses no fault data, so its value is unchanged by the source of the fault data.

The supervised SVM is the only detector that learns its decision boundary directly from fault examples, so it is the genuine test of substituting synthetic faults for real labels. The synthetic-trained SVM reaches a mean F_1 of 0.474 on the held-out channels (median 0.452), against 0.631 (median 0.779) for the real-trained SVM on those same channels, a retention of 75%. A paired Wilcoxon signed-rank test across the 20 channels gives $p = 0.13$: the difference is not statistically significant, so on this unseen-channel set the synthetic-trained detector is not detectably worse than the real-trained one. The paired difference itself is 0.157 (a 25% relative reduction), with a 95% confidence interval of $[-0.04, +0.35]$: the interval is wide and spans zero, so the data are consistent with anything from a slight synthetic advantage to a substantial real-data shortfall, and the non-significant result reflects the limited power of a twenty-channel comparison rather than demonstrated equivalence. This is the strongest defensible form of the substitution claim, and it is stated with the like-for-like held-out denominator rather than against the larger all-channel baseline of Chapter 3. The two unsupervised detectors depend on fault data only to set a threshold, and their deployable threshold is the classical $k = 3$ rule, computed from nominal training error and using no fault data at all. Their deployable performance is therefore unchanged by whether the fault data is real or synthetic: the VAE reaches 0.397 and the LSTM 0.543 in both cases (the small LSTM difference, 0.543 against 0.547, is statistically indistinguishable at $p = 0.70$). The substitution result is consequently meaningful where it is hardest, for the supervised detector that genuinely needs labelled faults, and trivially preserved for the detectors that do not. Figure 5.1 places the synthetic-trained detectors against their held-out real references.

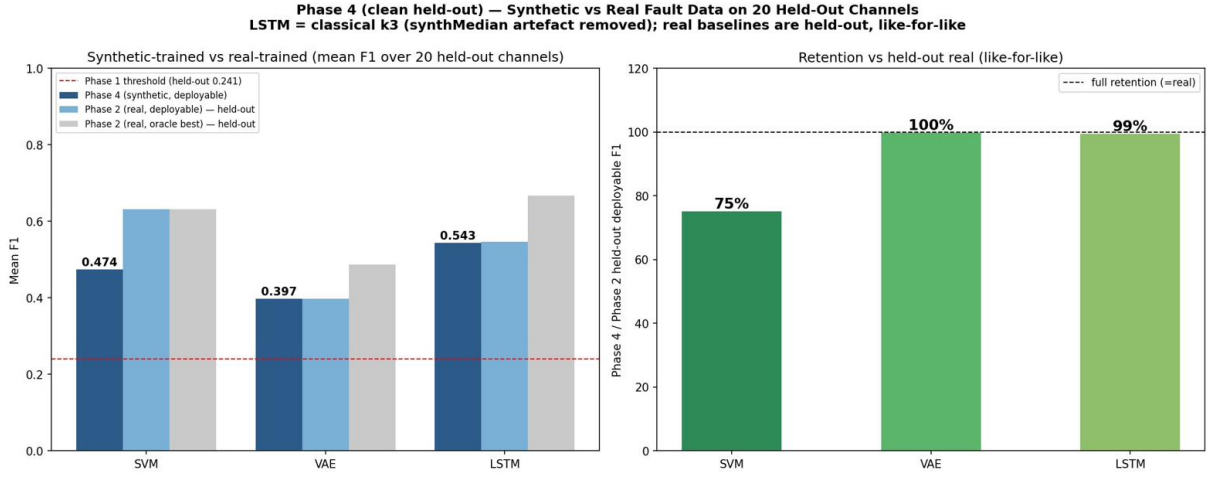


Figure 5.1: Phase 4 on the 20 held-out channels. Left: synthetic-trained against real-trained deployable mean F_1 for the three detectors, with the held-out threshold baseline for reference; for the unsupervised detectors the deployable rule is the classical $k = 3$. Right: retention of the synthetic-trained detector relative to the like-for-like held-out real reference. The supervised SVM retains 75% with no statistically significant loss ($p = 0.13$); the unsupervised detectors fall back to the parameter-free $k = 3$ rule, which synthetic data does not improve.

5.2. Campaign 2 — The Marginal Value of Data

The second campaign asks two complementary questions. Does adding a small number of *real* fault examples to synthetic training improve detection on unseen channels? And does adding *synthetic* examples to real training improve it? Under the clean protocol the added real windows are drawn exclusively from the 61 training channels, so both questions are genuinely about cross-channel generalisation.

5.2.1. Data efficiency: real windows from known channels do not transfer

Starting from synthetic-only training, real anomaly windows drawn from the training channels are added to the supervised classifier in increasing numbers, and detection is re-evaluated on the held-out channels. Table 5.2 reports the resulting curve, which is flat: mean F_1 moves from 0.474 with no real windows to 0.465 with the full training-channel pool, varying only between 0.454 and 0.474 across the entire sweep, as plotted in Figure 5.2.

Real anomaly windows added	Mean F_1	Median F_1
0 (synthetic only)	0.474	0.452
1	0.473	0.453
2	0.474	0.454
5	0.459	0.455
10	0.454	0.416
20	0.465	0.439
all	0.465	0.494

Table 5.2: Data-efficiency curve on the 20 held-out channels: mean and median F_1 as real anomaly windows from the training channels are added to synthetic training. The curve is flat.

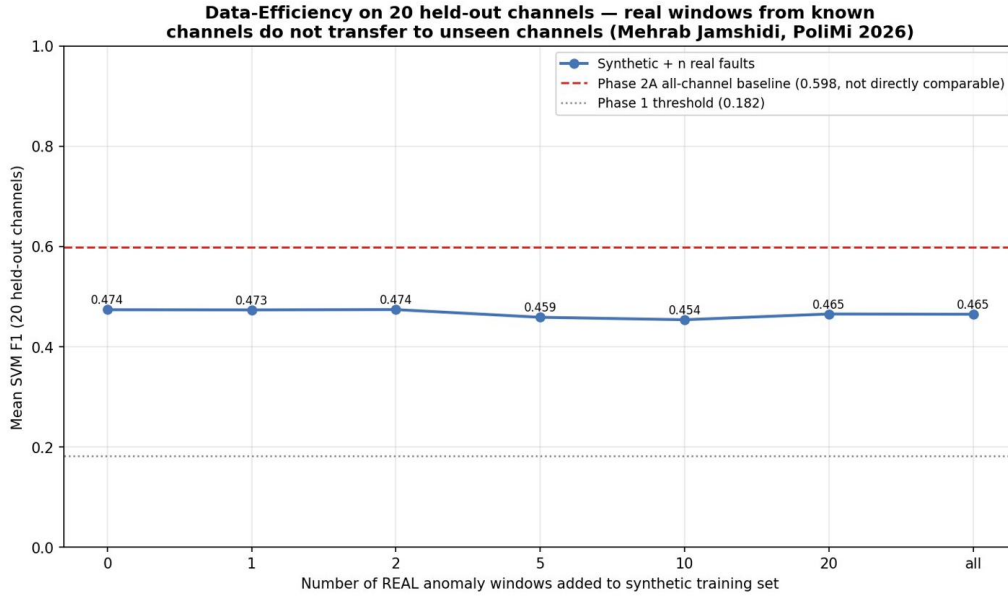


Figure 5.2: Data-efficiency on the 20 held-out channels. The curve is flat: real anomaly windows from known channels do not transfer to improve detection on unseen channels. The Phase 2 all-channel baseline (0.598) is shown for reference only and is not directly comparable, because it is measured on a different and larger set of channels.

An earlier, contaminated version of this experiment had reported a rising curve and concluded that one trained generator was worth approximately 11 to 13 real anomaly windows. That conclusion is withdrawn. It arose because the added real windows were

drawn from the same channels on which detection was being measured, so they were in-distribution for the evaluation set; under the clean protocol, where the added real windows come from different channels, they do not transfer, and the curve is flat. The corrected and more conservative finding is the honest one: real labelled anomaly windows from known channels do not improve detection on unseen channels, which is itself an informative statement about how poorly fault signatures generalise across telemetry channels. It also means that synthetic data and cross-channel real data carry comparable, modest transfer value, a comparison made explicit next.

5.2.2. Augmentation: synthetic substitutes for, but does not augment, real data

The complementary experiment fixes the available real data, now drawn from the training channels, and adds synthetic windows to it. On the held-out channels, training the supervised classifier on synthetic windows alone reaches a mean F_1 of 0.474, and training on real windows borrowed from the training channels reaches a plateau of 0.469: statistically the same value. Synthetic faults therefore transfer to unseen channels about as well as real faults transferred from other channels, and neither source is privileged. Adding synthetic windows on top of the cross-channel real data gives no improvement on the mean, as Table 5.3 and Figure 5.3 show: with 20 real windows present, the mean F_1 stays between 0.46 and 0.47 as the number of added synthetic windows is swept from 0 to 2,000.

Real windows	+0 synth	+500 synth	+1000 synth	+2000 synth
0 (synthetic only)	—	0.454	0.474	0.450
5	0.384	0.470	0.455	0.446
10	0.443	0.449	0.455	0.454
20	0.460	0.467	0.466	0.471
all	0.469	0.466	0.465	0.463

Table 5.3: Augmentation grid on the 20 held-out channels (supervised classifier, mean F_1): cross-channel real and synthetic windows reach the same plateau near 0.47, and adding synthetic windows on top of cross-channel real data gives no improvement on the mean. The “0 real, +0 synth” cell is omitted, as it leaves no training data.

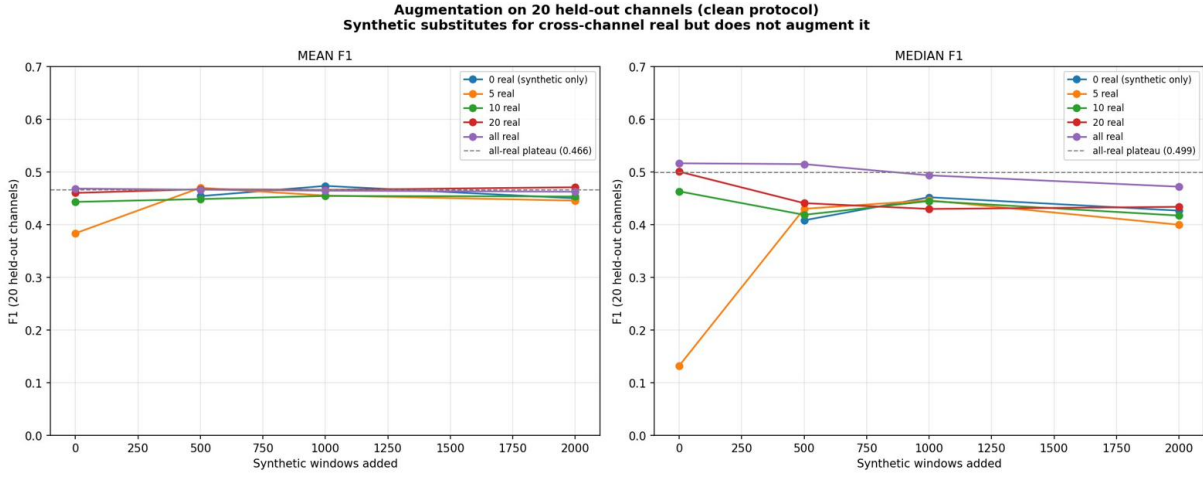


Figure 5.3: Augmentation on the 20 held-out channels under the clean protocol. Left (mean): synthetic-only training reaches the same plateau (about 0.47) as training on all cross-channel real windows, and adding real on top gives no further gain. Right (median): where cross-channel real data already yields a working detector, adding synthetic windows mildly degrades it, a weak residual distributional-drift signal. Synthetic substitutes for cross-channel real but does not augment it.

A weak residual signal does survive on the median: where the cross-channel real data already yields a working detector, adding synthetic windows mildly degrades it, with the median F_1 at 20 real windows falling from roughly 0.50 to 0.43 as synthetic windows are added. The conclusion is therefore twofold and measured. Synthetic faults are a viable substitute for real faults when no real labelled faults are available for a channel, the deployment-relevant case, but they are not a useful augmentation on top of real data, and mixing the two slightly hurts the median through distributional drift. This is reported as a negative result about augmentation, not as a recommendation against synthetic substitution; it is also milder and better posed than the contaminated all-channel version, in which the apparent strong degradation was largely an artefact of the added real windows being the channel’s own in-distribution test anomalies, which synthetic data could only dilute.

5.3. Campaign 3 — Label-Free Threshold Calibration

The unsupervised detectors of Chapter 3 left a gap between their deployable $k = 3$ performance and their oracle-threshold performance, the headroom recoverable only with

knowledge of the test labels. The third campaign asks whether synthetic data can close that gap by informing the threshold without recourse to real labels. Four synthetic-informed rules are tested against the classical $k = 3$ default and the oracle ceiling: a rule that picks the best fixed k from the synthetic score distribution (**bestK**), a rule that maximises an F_1 proxy on a synthetic-real mixture (**mixF1**), a rule that targets a synthetic recall level (**synthRecall**), and a rule that thresholds at the median synthetic score (**synthMedian**). Table 5.4 reports the outcome on the held-out channels.

Detector	$k = 3$	bestK	mixF1	synthRecall	synthMedian	Oracle
VAE	0.397	0.397	0.315	0.350	0.309	0.486
LSTM	0.543	0.545	0.588	0.549	0.595*	0.614

Table 5.4: Label-free threshold calibration on the 20 held-out channels (mean F_1). The oracle column requires the test labels and is an upper bound. * The apparent LSTM **synthMedian** win over $k = 3$ is a grid-resolution artefact, examined in Section 5.3.1.

For the VAE the result is clean and unambiguous: the deployable $k = 3$ rule reaches 0.397, exactly matched by **bestK** (0.397), and every genuinely synthetic-informed rule scores lower (0.315, 0.350, 0.309), well below the oracle ceiling of 0.486. A paired test of the best synthetic-informed rule against $k = 3$ gives $p = 0.29$: synthetic data does not produce a label-free threshold rule that beats the classical three-sigma default for the VAE. The headroom between $k = 3$ (0.397) and the oracle (0.486) is real, but it is governed by the *nominal* score distribution, which the detector already estimates well from its nominal training data, not by the fault distribution that synthetic data improves. That is why synthetic faults add no information the k -sigma rule does not already capture. This identifies label-free threshold calibration as a problem that generative augmentation does not solve, and that remains open for future work.

5.3.1. The LSTM **synthMedian** result is a grid-resolution artefact

For the LSTM, Table 5.4 appears to show a synthetic-informed rule beating the classical default: **synthMedian** reaches 0.595 against $k = 3$'s 0.543. This apparent reversal must not be read as evidence that synthetic calibration improves the LSTM, because it is a measurement artefact, and three independent checks establish that it is.

First, a paired Wilcoxon test of **synthMedian** against $k = 3$ across the 20 channels gives

$p = 0.69$, so the difference is not statistically significant. Second, **synthMedian** exceeds the within-run oracle ceiling on 7 of the 20 channels (G-1, G-2, T-2, R-1, F-3, T-5 and D-14); since the oracle selects the best threshold using the test labels, no label-free rule can legitimately exceed it, so these are impossible “wins” produced by the coarseness of the point-adjust score under a single rare positive. The clearest case is channel R-1, where **synthMedian** scores 1.000 against an oracle ceiling of 0.022, the entire “detection” resting on one fortuitously flagged step in a segment that point-adjust then credits in full. Third, on the median across channels, which is robust to exactly this kind of single-channel outlier, the ordering reverses, with $k = 3$ at 0.689 ahead of **synthMedian** at 0.663.

The LSTM deployable result is therefore reported as the classical $k = 3$ rule (0.543 mean, 0.689 median), and synthetic-informed rules are reported as not genuinely improving on it. This preserves the finding that matters operationally: the nominally trained LSTM with the classical $k = 3$ threshold remains the strong, recommended deployable detector, and synthetic threshold calibration does not improve it. For completeness, an ablation that trains a discriminative LSTM classifier directly on the raw synthetic time series, rather than scoring prediction error, reaches only 0.320 on the held-out channels, well below the engineered-feature SVM on the same synthetic data, confirming that a raw time-series classifier transfers less well than the feature-based classifier.

5.4. Campaign 4 — Sensitivity Analysis with a Controllable Fault Generator

The fourth campaign is the principal methodological contribution of the thesis, and the one that the clean protocol leaves entirely intact. The preceding campaigns ask whether synthetic data can replace real data; this campaign exploits a capability that real data does not possess at all. In a fixed labelled dataset, each fault occurs once, at a single uncontrollable combination of magnitude and duration, and the dataset cannot say how detection performance would change if the fault were milder, shorter, or more severe. A trained generator removes this limitation: it can synthesise faults at any chosen severity and duration, inject them into otherwise-nominal real telemetry, and so map detector capability across the entire operational parameter space. Under the clean protocol this sweep is run only on the 20 held-out channels, using synthetic anomalies from the generator that never saw them.

One point of interpretation should be stated at the outset. The surface characterises how the detector responds to faults produced by this generator and parameterised by the

injection model of Equation (5.1): severity and duration are controlled through the blend coefficient α and the window length d , not through a physical fault model. It is therefore a statement about detectability of *generator-produced* faults at a given (α, d) , and its extrapolation to an arbitrary real fault of the same nominal severity and duration rests on the realism of the generator established in Chapter 4; validating the surface against real graded faults is identified as future work in Chapter 6. With that scope fixed, the surface is exactly the design-time characterisation that a fixed labelled dataset cannot provide.

5.4.1. Injection methodology

For each held-out channel the LSTM, the strongest detector of Chapter 3, is trained on the channel’s nominal training data and its deployable $k = 3$ threshold is set. The longest contiguous nominal segment of the channel’s test split is extracted as a clean stream the detector has never seen. A synthetic fault is then injected into this stream as a severity-weighted blend of the nominal signal and a synthetic anomaly window g from the Phase 3 generator,

$$x_t^{\text{inj}} = (1 - \alpha) x_t^{\text{nom}} + \alpha g_{t-\tau}, \quad t \in [\tau, \tau + d), \quad (5.1)$$

in which τ is the injection onset, d the duration and α the severity: $\alpha = 1$ replaces the nominal signal entirely with the synthetic fault, $\alpha < 1$ blends the fault into the nominal signal, and $\alpha > 1$ amplifies the fault beyond full replacement. Six severity levels, $\alpha \in \{0.10, 0.25, 0.50, 0.75, 1.00, 1.50\}$, and five durations, $d \in \{16, 32, 64, 128, 256\}$ time steps, define a 6×5 grid of operating conditions. At each grid cell the injection is repeated over five replicates per channel with random onset positions and randomly selected synthetic windows, three non-overlapping injections per replicate, and the per-cell mean F_1 and detection latency are aggregated across the held-out channels. Detection latency is the number of time steps between onset τ and the first time step the detector flags.

5.4.2. The detection-performance surface

Table 5.5 reports the mean F_1 surface, with the no-injection baseline reported separately in Section 5.4.5. Performance rises monotonically with both severity and duration, from 0.255 at the smallest, shortest fault ($\alpha = 0.10$, $d = 16$) to 0.845 at the largest, longest ($\alpha = 1.50$, $d = 256$). The structure carries a clear operational message for this prediction-error detector: over the range examined, duration dominates severity. Holding severity at $\alpha = 1.0$, increasing duration from 16 to 256 time steps raises mean F_1 from 0.375 to 0.802,

a gain of 0.43; holding duration at $d = 256$, increasing severity from 0.10 to 1.50 raises mean F_1 from 0.694 to 0.845, a gain of only 0.15. A fault is easier to catch by lasting longer than by being more severe, which reflects that the LSTM accumulates evidence over time; the balance could differ for a detector with a different scoring mechanism, and the surface is therefore reported as a characterisation of this detector rather than a universal property of the fault classes.

$\alpha \setminus d$	16	32	64	128	256
0.10	0.255	0.336	0.480	0.613	0.694
0.25	0.276	0.369	0.522	0.654	0.742
0.50	0.328	0.415	0.558	0.672	0.791
0.75	0.346	0.456	0.563	0.691	0.793
1.00	0.375	0.494	0.591	0.708	0.802
1.50	0.441	0.561	0.667	0.772	0.845

Table 5.5: Detection-performance surface: mean F_1 of the LSTM detector across fault severity α and duration d (time steps), aggregated over the 20 held-out channels (19 at $d = 256$).

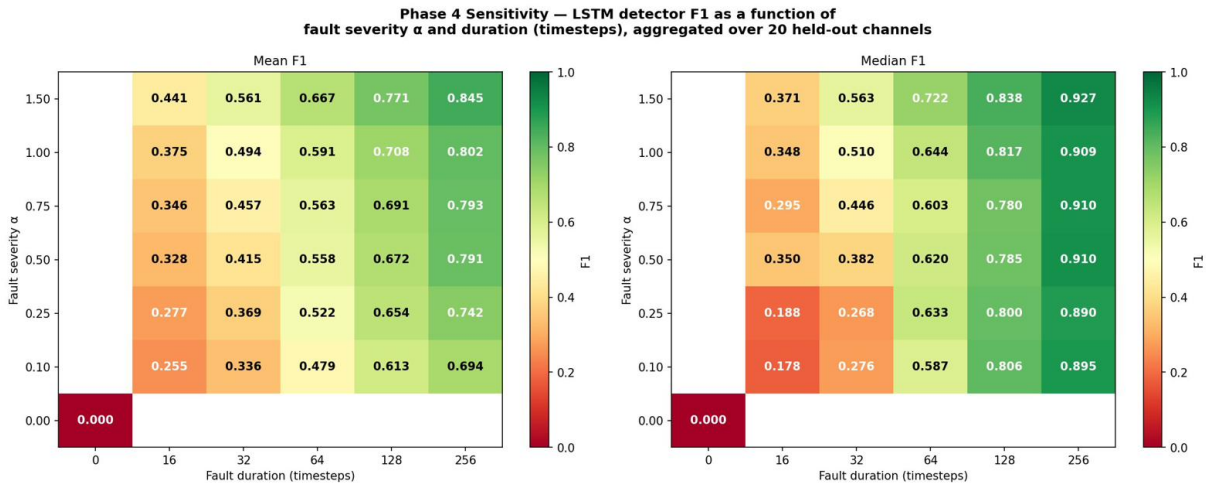


Figure 5.4: Detection-performance surface (mean and median F_1) over fault severity and duration, on the 20 held-out channels. Detectability increases monotonically with both axes; duration is the stronger determinant over the range examined. The $\alpha = 0$ column is the no-injection baseline of Section 5.4.5. This surface cannot be extracted from a fixed labelled dataset, in which each anomaly occurs at one uncontrollable operating point.

5.4.3. The detection-latency surface

Table 5.6 reports the companion latency surface, plotted in Figure 5.5. Latency falls steeply with severity: a severe short fault ($\alpha = 1.50$, $d = 16$) is detected within one time step on average, whereas a subtle long fault ($\alpha = 0.25$, $d = 256$) takes on average 30.7 time steps to flag. A useful operational anchor is the cell closest to a typical real fault, $\alpha = 1.0$ and $d = 64$, at which the mean F_1 is 0.591 and the mean latency is 3.30 time steps. The latency surface is conditional on detection, as explained in Section 5.4.5, and must be read together with the F_1 surface: together the two form a *partial FDIR design data sheet* that states, for any required detection latency and any minimum fault to be caught, the detection performance the LSTM would achieve, a characterisation that no fixed labelled dataset can provide.

$\alpha \setminus d$	16	32	64	128	256
0.10	3.2	6.0	12.6	17.0	27.4
0.25	2.3	5.0	13.9	19.7	30.7
0.50	1.6	3.1	7.3	14.4	26.3
0.75	0.9	2.6	3.5	11.3	19.9
1.00	1.0	1.7	3.3	8.1	20.0
1.50	0.8	1.8	2.4	7.1	10.2

Table 5.6: Detection-latency surface: mean number of time steps from fault onset to first detection, across severity α and duration d , over the 20 held-out channels. The average at each cell is taken only over injections the detector actually flagged (Section 5.4.5).

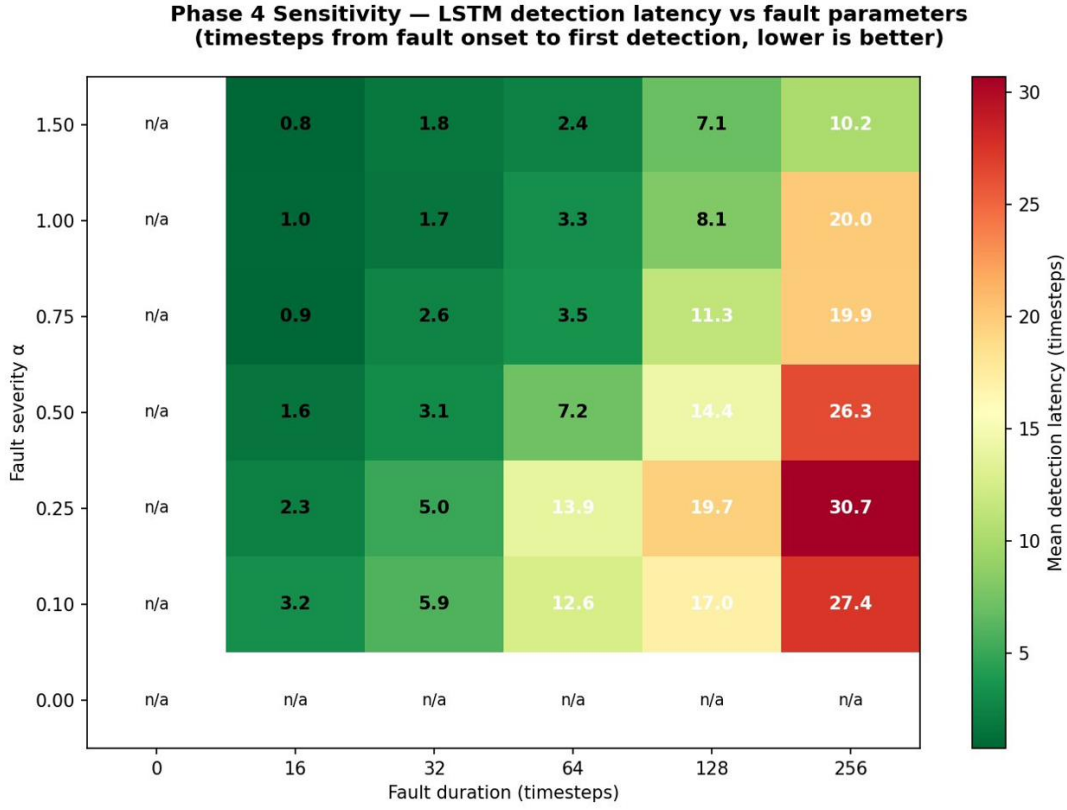


Figure 5.5: Detection-latency surface (mean time steps to first detection) over fault severity and duration, on the 20 held-out channels; lower is better. Latency is computed only over detected injections and must be read together with the F_1 surface of Figure 5.4.

5.4.4. Per-class behaviour

Disaggregating the surface by anomaly class reveals two qualitatively different detector responses. On point-anomaly channels the surface is almost flat in severity and rises essentially only with duration: at $d = 256$, increasing severity from 0.10 to 1.50 changes the point-class F_1 by only +0.08. On contextual-anomaly channels severity matters strongly: the same increase raises the contextual-class F_1 by +0.35, from 0.59 to 0.94, because a more severe contextual injection produces a larger departure from the nominal autocorrelation structure that the LSTM keys on. The controllable generator thus exposes not only how detectable a fault is, but which property of the fault, its amplitude or its duration, governs detectability for each fault family, a distinction a fixed dataset cannot draw. The full per-class F_1 and latency surfaces are reported in Chapter C.

5.4.5. Robustness of the sensitivity surface

Three elements guard the interpretation of the surfaces. First, a no-injection false-positive baseline is measured by evaluating the detector on each clean stream with no fault injected, so that the ground truth is entirely nominal and the result isolates false-positive behaviour. The median false-positive rate across the un-injected streams is low, at 0.011, confirming that the surface is not contaminated by a baseline tendency to over-flag; the mean is higher, at 0.123, inflated by a small number of channels (notably D-5 and R-1) whose clean stream is flagged almost in its entirety, as the per-channel rates in Table C.5 make explicit. Both are reported so that the floor is not understated. Second, the number of channels contributing to each grid cell is recorded explicitly: each cell aggregates over all 20 held-out channels for durations up to 128, and over 19 channels at $d = 256$, because one channel's clean nominal segment is too short to host the longest injection. No cell's mean is therefore computed over a silently different sample than its neighbours. Third, as already noted, the latency average at each cell excludes undetected injections: it reports response speed conditional on detection, and undetected injections lower recall and F_1 but do not enter the latency average. The latency surface must accordingly be read jointly with the F_1 surface, never on its own.

5.4.6. Channel P-1 as a genuine unseen-channel result

Because channel P-1 falls in the held-out set, its Phase 4 results are now clean unseen-channel results, and they make the central argument of the thesis in miniature. P-1's three anomaly windows are contextual and never exceed the nominal amplitude bounds, so classical threshold detection is essentially blind to it, scoring $F_1 = 0.011$. On this channel the synthetic-trained SVM reaches 0.169 and the VAE at $k = 3$ reaches 0.000, but the nominally trained, $k = 3$ -thresholded LSTM recovers a mean F_1 of 0.718: meaningful detection on a contextual channel from which the generator was never shown a fault, and effectively identical to the 0.726 the same detector achieves when P-1 is included in training, exactly as expected for a detector that learns only from nominal data. The Phase 2 real-data LSTM at its oracle threshold reaches 0.814 on this channel. The progression from 0.011 to 0.718 on a fault class that classical FDIR cannot register, obtained on a genuinely unseen channel, is a more honest and more compelling illustration than the contaminated version it replaces. Figure 5.6 shows P-1 across all phases.

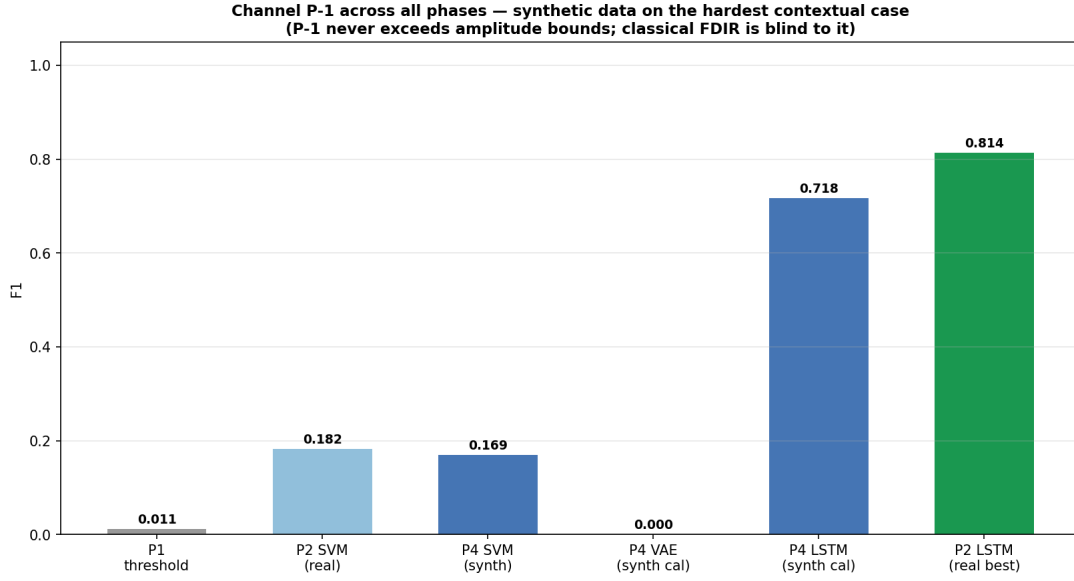


Figure 5.6: Channel P-1 across all phases, as a held-out unseen-channel result. P-1 is a contextual anomaly that never exceeds the nominal amplitude bounds, so classical FDIR is blind to it ($F_1 = 0.011$). The nominally trained, $k = 3$ -thresholded LSTM recovers $F_1 = 0.718$ on this unseen channel, against the Phase 2 real-data LSTM oracle of 0.814.

5.5. Statistical Summary

Every synthetic-versus-real comparison reported in this chapter was tested with a paired Wilcoxon signed-rank test over the 20 held-out channels; Table 5.7 consolidates them. With $n = 20$ the tests are not highly powered, so a non-significant result is reported as the absence of a detectable difference rather than as proof of equality.

Comparison	Synthetic	Real / ref.	p	Verdict
SVM synthetic vs. real (substitution)	0.474	0.631	0.13	no significant difference
LSTM $k=3$ synthetic vs. real	0.543	0.547	0.70	indistinguishable
LSTM synthMedian vs. $k=3$	0.595	0.543	0.69	not significant (artefact)
VAE best synthetic rule vs. $k=3$	0.350	0.397	0.29	$k=3$ not beaten

Table 5.7: Paired Wilcoxon signed-rank tests on the 20 held-out channels. “No significant difference” denotes failure to reject equality at the 0.05 level.

5.6. Discussion

Read under the clean protocol, the four campaigns separate cleanly into a measured substitution result and an unqualified methodological one. The substitution result is real but modest, and it is now properly hedged. Synthetic fault data substitutes for real labelled faults for the one detector that genuinely depends on them, the supervised classifier, which on unseen channels is not significantly worse trained on synthetic data than on real data (0.474 against 0.631; $p = 0.13$). For the unsupervised detectors the deployable threshold is the parameter-free $k = 3$ rule, which uses no fault data and which synthetic-informed calibration does not beat, so synthetic data changes their deployable performance not at all. The two negative findings sharpen rather than weaken this picture: synthetic augmentation does not improve on cross-channel real data and mildly degrades the median through distributional drift, and the data-efficiency curve is flat because fault signatures do not transfer well across channels. Both follow from the same underlying fact, that the quantities those experiments depend on, the nominal score distribution and cross-channel generalisation, are not the quantities that synthetic fault generation improves.

The honest operational recommendation that follows is correspondingly narrow, and therefore trustworthy: a generator trained on a spacecraft’s historical fault record is worth deploying to train a supervised detector for fault types too rare to have accumulated sufficient real examples, and it is not worth deploying to augment an already-adequate real dataset or to calibrate an unsupervised detector’s threshold, where the classical $k = 3$ rule remains the strong default.

Against this measured assessment of substitution stands the unqualified contribution of the fourth campaign, which the held-out protocol leaves entirely intact because it never compares synthetic data against real data at all. The sensitivity analysis demonstrates a use of generative models for FDIR that does something real data cannot: it characterises a detector’s capability as a continuous function of fault severity and duration, on channels the generator never saw, producing a design data sheet that states the minimum detectable fault and the detection latency across the operational envelope, and that shows fault duration to matter more than fault severity for detectability. This reframes the role of the generative model from a data-augmentation tool, whose value is conditional and modest, to an FDIR design and verification instrument, whose value is unique. It is in this second role, as a controllable testbed for characterising and certifying detection logic before flight, that the generative model, understood as the learning core of a prospective Cognitive Digital Twin rather than as a twin in itself, most clearly supports FDIR design

and testing, and it is the foundation on which the conclusions of Chapter 6 build.

6 | Conclusions and Future Developments

This thesis set out from a structural limitation of classical spacecraft Fault Detection, Isolation and Recovery. A fault that accumulates gradually, or that alters the temporal pattern of a signal without breaching its amplitude bounds, produces no threshold crossing and is never flagged. In the public record of real in-orbit anomalies, 43 of 105 labelled events are of exactly this contextual type. The controlled attitude-dynamics simulation of Section 2.7 confirmed the failure in a spacecraft-engineering setting: a gyroscope drift fault drove the estimation error to 14 deg/s while the classical variance monitor never triggered. The thesis asked whether generative models trained on real spacecraft telemetry could close that gap, and how far synthetic fault data could substitute for, augment, and characterise the detectors that an operational FDIR system would deploy. It answered across four experimental phases, all evaluated on the same 81 NASA SMAP/MSL channels under a single point-adjust protocol, so that every result is directly comparable.

6.1. Summary of Findings

The detection phases established that data-driven detectors trained on real telemetry close the contextual-anomaly gap by a wide margin. The classical threshold baseline achieved a mean F_1 of only 0.183, collapsing to 0.052 on the contextual channels. Against that floor, the unsupervised Variational Autoencoder reached 0.590 overall and 0.638 on the contextual channels at the oracle threshold and trained without labelled fault data, and the LSTM regression detector reached 0.743 overall and 0.836 on the contextual channels, also at the oracle threshold. The latter is a sixteenfold improvement, under point-adjust, over the classical detector on the very fault class it cannot reach. The LSTM's agreement with the published benchmark of Hundman et al. [15], to within 0.006 on the SMAP channels, independently validated the entire pipeline.

The generative phase showed that a conditional Wasserstein GAN, equipped with the four architectural contributions of Section 4.2 and trained on the real anomaly windows of the

61 training channels held aside under a clean protocol, produces synthetic fault data of measurable quality. The contextual class reached a Fréchet Distance of 0.141, with the real autocorrelation structure matched closely for both classes, and a classifier trained purely on synthetic data was, on unseen held-out channels, not statistically distinguishable from one trained on real data (0.474 against 0.631; Wilcoxon $p = 0.13$ at $n = 20$, a non-detectable difference rather than a proven equivalence). The autocorrelation-matching term, the most direct of the four contributions, drove the temporal-structure error to near zero within a hundred training steps. Because that decay is partly expected by construction, and because the contributions were not isolated in a controlled ablation, this is reported as consistent with the term’s role rather than as proof of it.

The transfer phase, run entirely on the held-out channels, split the central question into a measured result and a sharper account of where synthetic data applies. Synthetic fault data substitutes for real labelled faults for the one detector that genuinely depends on them: on unseen channels the supervised classifier trained on synthetic data is not statistically worse than one trained on real data (0.474 against 0.631; $p = 0.13$). The unsupervised VAE and LSTM learn mainly from nominal data, and their deployable threshold is the parameter-free $k = 3$ rule, which uses no fault data and which synthetic-informed calibration does not beat; synthetic data therefore leaves their deployable performance unchanged. Two negative findings tighten this picture rather than weaken it. Synthetic augmentation does not improve on cross-channel real data and mildly degrades the median through distributional drift, and the data-efficiency curve is flat because real fault windows do not transfer well across channels. Both follow from the same fact: the quantities those experiments depend on are not the quantities that synthetic fault generation improves. The earlier claim that one generator was worth roughly 11 to 13 real anomaly windows is withdrawn, since it rested on real windows drawn from the channels under evaluation.

6.2. Answer to the Research Question

The research question admits a precise, two-part answer. Generative models trained on real spacecraft telemetry do improve fault detection over classical FDIR, and they do so in two distinct roles that should not be conflated.

In the first role, as detectors, the reconstruction-based VAE and the prediction-based LSTM close the contextual-anomaly gap that classical FDIR cannot reach, the LSTM most strongly, while learning autonomously from nominal data. In the second role, as a source of synthetic fault data, the conditional GAN substitutes for real labelled data

without statistically significant loss on unseen channels, but only for the supervised detector that genuinely depends on the fault distribution. For the unsupervised detectors the parameter-free $k = 3$ rule remains the deployable default, and synthetic data neither augments an adequate real dataset nor calibrates an unsupervised threshold. The clearest value of the generative model emerged not as a competitor to real data but in a capability real data lacks. As a controllable fault generator, it characterised detector performance, on channels it had never seen, as a continuous function of fault severity and duration. The result is a partial FDIR design data sheet, stating the minimum detectable fault and the detection latency across the operational envelope and showing fault duration to matter more than fault severity for detectability. No fixed labelled dataset can provide this, because each real fault occurs at a single uncontrollable operating point.

The operational recommendation that follows is narrow, and therefore trustworthy. A generator trained on a spacecraft’s historical fault record is worth deploying for two things: to train a supervised detector for fault types too rare to have accumulated enough real examples, and as a controllable testbed for characterising and certifying detection logic before flight. It is not worth deploying to augment an already-adequate real dataset, nor to calibrate an unsupervised detector’s threshold. Mapped onto the Digital Twin maturity framework of Wei et al. [31], the resulting system moves spacecraft FDIR from its present state of the art, machine-learning detection alongside classical monitoring at the Prescriptive level, toward the Cognitive level, at which the twin learns and continually characterises its own fault-detection capability from real operational data.

6.3. Limitations

Five limitations bound these conclusions and motivate the developments that follow. First, all detectors and generators operate on the primary sensor measurement alone; the correlated context channels of the dataset are unused, so faults whose signature spans multiple sensors are not exploited. Second, the study uses the NASA SMAP/MSL benchmark, whose clean train–test split and moderate scale made it the right basis for developing the methodology, but which is smaller and less operationally demanding than the ESA benchmark of Kotowski et al. [21]. Third, the deployable-threshold problem remains open: the gap between the $k = 3$ and oracle thresholds is real headroom that neither the classical rule nor the synthetic-informed rules tested here can recover without real labels. Fourth, the synthetic-to-real transfer results rest on a held-out set of 20 channels. This is the most defensible design available on a dataset in which most channels carry a single labelled anomaly, but it limits the statistical power of the comparisons, so non-significant

differences are reported as the absence of a detectable effect rather than as equality; it also exposes a genuine property of the data, that fault signatures generalise only weakly across telemetry channels. Fifth, the sensitivity surface of Section 5.4 characterises detectability of faults produced by the generator and parameterised by the injection model of Equation (5.1), not of arbitrary physical faults; extrapolating it to real graded faults of matched severity and duration rests on the realism of the generator and remains to be validated against real fault data.

6.4. Future Developments

Several directions extend the work along clear lines.

Multivariate detection and generation. The highest-priority extension is to bring in the correlated telemetry channels alongside the primary measurement. A multivariate VAE or LSTM would detect faults whose signature is a change in the relationship between channels rather than in any single one, and a multivariate generator would synthesise jointly consistent fault windows. This should help contextual detection most, since cross-channel correlation often carries the fault signature most clearly.

Scaling to operational benchmarks. Validating the methodology on the ESA Anomaly Detection Benchmark would test it against operational realism: irregular sampling, communication gaps, anomalies present in training data, and a scale exceeding 700 million points per mission, as well as the superior label quality of its two-stage expert annotation. This is the natural path from a tractable, well-benchmarked problem toward industrial deployment.

Stronger detector and generator architectures. The Anomaly Transformer of Xu et al. [32], which reports F_1 above 0.90 on these datasets, sets the upper-bound reference that the detectors here did not reach; a self-attention detector would narrow that gap. On the generative side, diffusion models [22, 29] offer higher fidelity than GANs, and their sampling cost is acceptable for the offline, ground-based generation this application needs. Finer sub-type conditioning of the point-anomaly generator would also address the residual point-class fidelity gap identified in Chapter 4. Finally, a controlled ablation isolating the marginal contribution of each of the four architectural choices, together with an explicit nearest-neighbour memorisation check, would strengthen the evidence that the generator’s quality derives from those choices rather than from the adversarial objective alone.

Validating the sensitivity surface. The design data sheet from the controllable-fault sensitivity analysis characterises detectability of generator-produced faults. Confirming that it predicts detection of real graded faults, for instance by comparing the surface against real anomalies binned by estimated severity and duration or against hardware-in-the-loop fault injections, would turn it from a characterisation of the detector’s response to the generator into a validated FDIR design tool.

Label-free threshold calibration. The negative calibration result points to a specific open problem: recovering the oracle-threshold headroom without real labels. Because that headroom is governed by the nominal score distribution, future work should target richer nonparametric models of that distribution, such as the dynamic thresholding of Hundman et al. [15], rather than the fault distribution that synthetic data improves.

Toward a closed-loop Cognitive Digital Twin. The components developed here, unsupervised detectors that learn from nominal data, a controllable fault generator, and a sensitivity-characterisation procedure, are the building blocks of a closed-loop Cognitive Digital Twin for FDIR. Integrating them, so that the twin continuously updates its detectors from incoming nominal telemetry, re-characterises its detection capability as the spacecraft ages, and synthesises rehearsal faults for onboard verification, would realise the Level-5 maturity toward which this thesis points, and would extend the framework of Wei et al. [31] from launch-vehicle electronics to autonomous spacecraft fault management.

6.5. Closing Remark

The contribution of this thesis is twofold. It establishes, on real spacecraft telemetry and against a published benchmark, that data-driven detectors close the contextual-anomaly gap left open by classical FDIR. And it shows that a generative model of spacecraft faults, though only a bounded substitute for real training data, becomes a uniquely valuable instrument when used not to replace data but to characterise detection capability across the operational envelope. It is in this second role, as a controllable design and verification testbed for fault-detection logic, that generative Digital Twin methods most clearly support FDIR design and testing, and it is where their further development is most likely to repay the effort.

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A | Implementation Details and Hyperparameters

This appendix collects the full architecture and training specifications of the four models used in the thesis, supplementing the descriptions in Chapter 3 and Chapter 4. All models share the preprocessing and reproducibility conventions of Section 2.4: a window of 64 time steps, channel-level z-score normalisation, and the fixed random seed 42. All models were implemented in PyTorch.

A.1. Supervised Classifier (Phase 2, Step A)

The supervised classifier operates on six time-domain features extracted from each 64-step window: the window mean, standard deviation, minimum, maximum, range (maximum minus minimum), and the lag-one autocorrelation, computed as the Pearson correlation between the window and itself shifted by one time step. The first five summarise the amplitude distribution; the lag-one autocorrelation captures the temporal-pattern change that characterises contextual anomalies. Table A.1 gives the classifier configuration. Evaluation uses a leave-one-out cross-validation protocol over the anomalous windows of each channel, which guarantees a class-balanced evaluation fold for every channel regardless of how few anomalies it contains; predictions are accumulated across folds and precision, recall and F_1 computed on the pooled result.

Component	Specification
Input features	6: mean, std, min, max, range, lag-1 autocorrelation
SVM	RBF kernel, $C = 10$, balanced class weights
MLP (ANN)	two hidden layers (32 and 16 units), ReLU activations
MLP optimiser	Adam, lr = 0.001, L_2 weight = 0.001
MLP early stopping	patience 20 epochs on 15% held-out validation
Evaluation	leave-one-out cross-validation per channel

Table A.1: Supervised classifier (SVM and MLP) configuration.

A.2. Variational Autoencoder (Phase 2, Step B)

The Variational Autoencoder is a symmetric fully connected network with an eight-dimensional latent space (an 8:1 compression of the 64-step window). Training through the stochastic sampling step uses the reparameterisation trick, $z = \mu + \epsilon \odot \exp(\frac{1}{2} \log \sigma^2)$ with $\epsilon \sim \mathcal{N}(0, I)$, and the objective is the evidence lower bound of Equation (3.2). The network is trained only on the nominal training split of each channel. Table A.2 gives the full specification.

Component	Specification
Input	64 normalised time steps (window of primary sensor signal)
Encoder	Linear($64 \rightarrow 32$) $\rightarrow$ ReLU $\rightarrow$ Linear($32 \rightarrow 16$) $\rightarrow$ ReLU $\rightarrow$ Linear($16 \rightarrow 8$) for μ and for $\log \sigma^2$
Latent dimension	8 (compression ratio 8:1)
Decoder	Linear($8 \rightarrow 16$) $\rightarrow$ ReLU $\rightarrow$ Linear($16 \rightarrow 32$) $\rightarrow$ ReLU $\rightarrow$ Linear($32 \rightarrow 64$)
Training loss	$\mathcal{L} = \text{MSE}(x, \hat{x}) + \beta \text{KL}[\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, I)]$, $\beta = 1.0$
Anomaly score	per-window reconstruction error, max-pooled to per-time-step
Optimiser	Adam, lr = 0.001; early stopping patience 20 epochs
Mean training epochs	57.4 (range 21–100; 59 of 82 channels stopped early)

Table A.2: Variational Autoencoder architecture and training configuration.

A.3. LSTM Regression Detector (Phase 2, Step C)

The LSTM regression detector predicts the next telemetry value from the preceding 64 steps and scores anomalies by the one-step-ahead squared prediction error of Equation (3.3). Training sequences are constructed with a stride of 8 (input `signal[t:t+64]`, target `signal[t+64]`), which reduces the number of training sequences by a factor of eight relative to unit stride — from roughly 2,800 to 350 per channel — without measurable loss of accuracy. Table A.3 gives the full specification.

Component	Specification
Input	sequence of 64 normalised time steps, shape (batch, 64, 1)
LSTM layers	2 stacked layers $\times$ 64 hidden units, <code>batch_first</code>
Output layer	Linear($64 \rightarrow 1$) — predicts the next single time step
Training loss	$\text{MSE}(\hat{x}_{t+1}, x_{t+1})$
Anomaly score	per-time-step squared prediction error
Training stride	8
Optimiser	Adam, lr = 0.001; gradient clipping max-norm = 1.0; patience 20
Mean training epochs	53.2 (range 21–100; 70 of 82 channels stopped early)

Table A.3: LSTM regression detector architecture and training configuration.

A.4. Conditional WGAN-GP (Phase 3)

This section gives the layer-by-layer architecture of the generator and critic deferred from Section 4.3; the training configuration is given in Table 4.1. Two generators are trained, one per anomaly class, under identical settings. Table A.4 specifies the generator and Table A.5 the critic. The generator contains 184,865 parameters and the critic 42,337.

Layer	Shape (in $\rightarrow$ out)	Activation
Input (noise z)	$(B, 100)$, $z \sim \mathcal{N}(0, I)$	—
Reshape	$(B, 100) \rightarrow (B, 100, 1)$	—
ConvTranspose1d	$100 \rightarrow 128$ ch, $k=8$, $s=4$	BatchNorm + LeakyReLU(0.2)
ConvTranspose1d	$128 \rightarrow 64$ ch, $k=8$, $s=4$	BatchNorm + LeakyReLU(0.2)
ConvTranspose1d	$64 \rightarrow 32$ ch, $k=8$, $s=4$	BatchNorm + LeakyReLU(0.2)
Conv1d	$32 \rightarrow 1$ ch, $k=3$	Tanh
Output	$(B, 64)$ in $[-1, +1]$	—

Table A.4: Generator architecture (per-class). Total parameters: 184,865.

Layer	Shape (in $\rightarrow$ out)	Activation
Input (window)	$(B, 64) \rightarrow (B, 1, 64)$	—
SN-Conv1d	$1 \rightarrow 32$ ch, $k=4$, $s=2$	LeakyReLU(0.2)
SN-Conv1d	$32 \rightarrow 64$ ch, $k=4$, $s=2$	LeakyReLU(0.2)
SN-Conv1d	$64 \rightarrow 128$ ch, $k=4$, $s=2$	LeakyReLU(0.2)
Flatten	$(B, 128, 8) \rightarrow (B, 1024)$	—
SN-Linear	$1024 \rightarrow 1$	none (Wasserstein score)
Output	$(B, 1)$, unbounded	—

Table A.5: Critic architecture. Spectral normalization (SN) is applied to every learnable layer; no batch normalization is used. Total parameters: 42,337.

B | Per-Channel Detection Results

This appendix reports the point-adjust F_1 score of each method on every channel, supplementing the aggregate results of Chapters 3 to 5. Two tables are given, reflecting the two evaluation protocols of the thesis. Table B.1 reports the Phase 1 and Phase 2 detection results on all 81 unique telemetry channels: **Thr.** is the z-score threshold baseline, and **SVM**, **VAE** and **LSTM** are the Phase 2 detectors on real data, with the VAE and LSTM at the oracle threshold. Table B.2 reports the Phase 3 and Phase 4 synthetic-to-real results on the 20 held-out channels, evaluated under the clean protocol of Section 2.5: **SVM (synth)** is the classifier trained on synthetic data alone, **VAE** and **LSTM** are the unsupervised detectors at the deployable $k = 3$ threshold, and **TSTR** is the train-on-synthetic, test-on-real classifier. The 20 held-out channels are marked with a dagger ($\dagger$) in Table B.1. The **Type** column marks each channel by its primary class, contextual (C) or point (P). The final row of each table gives the column means.

Table B.1: Per-channel point-adjust F_1 for the Phase 1–2 detectors on all 81 unique SMAP/MSL channels. Channels held out from the generative experiments are marked $\dagger$.

Channel	S/C	Type	Thr.	SVM	VAE	LSTM
C-1 $\dagger$	MSL	P	0.000	0.732	0.752	0.990
C-2	MSL	P	0.146	1.000	0.138	0.143
D-14 $\dagger$	MSL	P	0.916	0.974	0.710	0.868
D-15	MSL	P	0.000	0.987	0.952	0.996
D-16	MSL	C	0.000	0.952	1.000	1.000
F-4	MSL	P	0.156	0.833	0.156	0.886
F-5	MSL	P	0.712	0.667	0.920	0.984
F-7	MSL	C	0.000	0.690	0.845	0.816

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Table B.1 – *continued from previous page*

Channel	S/C	Type	Thr.	SVM	VAE	LSTM
F-8	MSL	C	0.129	0.183	1.000	0.970
M-1	MSL	C	0.000	0.335	0.779	0.990
M-2 [†]	MSL	C	0.697	0.957	0.852	0.992
M-3	MSL	C	0.000	0.142	0.502	0.984
M-4	MSL	C	0.000	0.348	0.772	0.929
M-5	MSL	C	0.000	0.872	0.510	0.979
M-6	MSL	P	0.295	1.000	0.284	0.295
M-7	MSL	P	0.641	0.800	0.562	0.985
P-10	MSL	P	0.127	0.952	0.156	0.158
P-11	MSL	P	0.433	0.895	0.657	0.828
P-14	MSL	P	0.057	0.923	0.057	0.058
P-15	MSL	P	0.013	0.667	0.017	0.027
S-2	MSL	P	0.462	1.000	0.164	0.333
T-12	MSL	C	0.000	0.308	0.702	0.984
T-13	MSL	C	0.000	0.800	0.221	0.940
T-4	MSL	P	0.111	0.261	0.756	0.800
T-5 [†]	MSL	P	0.333	1.000	0.327	0.485
T-8	MSL	C	0.096	0.312	0.870	0.917
T-9	MSL	P	0.347	1.000	0.485	0.576
A-1	SMAP	P	0.019	0.167	0.857	0.019
A-2	SMAP	C	0.000	0.254	0.629	0.528
A-3	SMAP	C	0.000	0.250	0.939	0.894
A-4	SMAP	C	0.000	0.424	0.775	0.913
A-5	SMAP	P	0.036	1.000	0.021	0.041
A-6	SMAP	P	0.033	0.196	0.018	0.941
A-7	SMAP	C	0.000	0.285	0.668	0.935
A-8 [†]	SMAP	C	0.000	0.167	0.000	0.971
A-9	SMAP	C	0.000	0.998	1.000	0.636
B-1 [†]	SMAP	P	0.035	0.933	0.045	0.046

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Table B.1 – *continued from previous page*

Channel	S/C	Type	Thr.	SVM	VAE	LSTM
D-1	SMAP	P	0.939	0.983	0.789	0.994
D-11	SMAP	P	0.000	0.833	0.037	1.000
D-12	SMAP	P	0.969	0.973	1.000	1.000
D-13	SMAP	P	0.101	0.960	0.115	0.116
D-2	SMAP	P	0.949	0.949	0.999	0.998
D-3	SMAP	P	0.965	0.985	0.931	0.908
D-4 [†]	SMAP	P	0.998	0.998	0.947	0.999
D-5 [†]	SMAP	P	0.012	0.164	0.013	0.013
D-6	SMAP	P	0.000	0.167	0.021	1.000
D-7	SMAP	P	0.981	0.985	1.000	1.000
D-8	SMAP	P	0.039	0.800	0.617	0.935
D-9 [†]	SMAP	P	0.884	0.899	1.000	1.000
E-1	SMAP	C	0.000	0.213	0.376	0.945
E-10 [†]	SMAP	C	0.000	0.826	0.912	0.874
E-11	SMAP	C	0.000	0.864	0.884	0.987
E-12	SMAP	C	0.000	0.216	0.615	0.956
E-13	SMAP	C	0.000	0.156	0.377	0.773
E-2	SMAP	P	0.000	0.158	1.000	1.000
E-3	SMAP	P	0.000	0.161	0.971	1.000
E-4	SMAP	P	0.000	0.159	0.872	0.990
E-5	SMAP	P	0.000	0.167	0.000	1.000
E-6	SMAP	P	0.312	0.293	0.674	0.963
E-7	SMAP	P	0.240	0.800	0.986	0.864
E-8 [†]	SMAP	P	0.000	0.159	1.000	0.637
E-9	SMAP	P	0.000	0.142	0.535	0.989
F-1	SMAP	P	0.015	0.471	0.000	0.667
F-2	SMAP	P	0.022	0.182	1.000	0.993
F-3 [†]	SMAP	C	0.000	0.167	0.000	0.030
G-1 [†]	SMAP	C	0.000	0.952	0.000	0.886

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Table B.1 – *continued from previous page*

Channel	S/C	Type	Thr.	SVM	VAE	LSTM
G-2 [†]	SMAP	P	0.049	0.889	0.526	0.748
G-3 [†]	SMAP	P	0.039	0.727	0.365	1.000
G-4 [†]	SMAP	P	0.125	0.615	0.422	0.405
G-6	SMAP	P	0.000	1.000	0.238	0.031
G-7	SMAP	C	0.458	0.917	0.778	0.918
P-1 [†]	SMAP	C	0.011	0.182	0.275	0.814
P-2 [†]	SMAP	P	0.693	1.000	0.579	0.944
P-3	SMAP	P	0.057	0.163	1.000	1.000
P-4	SMAP	P	0.066	0.690	0.915	0.942
P-7	SMAP	C	0.017	0.627	1.000	0.992
R-1 [†]	SMAP	P	0.022	0.167	0.022	0.022
S-1	SMAP	P	0.000	0.784	0.996	0.851
T-1	SMAP	P	0.000	0.124	0.987	0.405
T-2 [†]	SMAP	P	0.000	0.110	1.000	0.611
T-3	SMAP	P	0.022	0.405	0.897	0.140
Mean			0.182	0.598	0.590	0.743

Table B.2: Per-channel point-adjust F_1 for the Phase 3–4 synthetic-to-real experiments on the 20 held-out channels. SVM (synth) is trained on synthetic data alone; VAE and LSTM are at the deployable $k = 3$ threshold.

Channel	S/C	Type	SVM (synth)	VAE ($k=3$)	LSTM ($k=3$)	TSTR
C-1	MSL	P	0.222	0.752	0.990	0.270
D-14	MSL	P	0.974	0.710	0.905	0.974
M-2	MSL	C	0.027	0.816	0.993	0.076
T-5	MSL	P	1.000	0.327	0.676	1.000
A-8	SMAP	C	0.559	0.000	0.000	0.613
B-1	SMAP	P	0.072	0.045	0.046	0.072
D-4	SMAP	P	1.000	0.943	0.998	0.998

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Table B.2 – *continued from previous page*

Channel	S/C	Type	SVM (synth)	VAE ($k=3$)	LSTM ($k=3$)	TSTR
D-5	SMAP	P	0.286	0.013	0.013	0.025
D-9	SMAP	P	0.899	1.000	1.000	0.899
E-10	SMAP	C	0.650	0.888	0.817	0.683
E-8	SMAP	P	0.221	0.951	0.441	0.456
F-3	SMAP	C	0.000	0.000	0.020	0.000
G-1	SMAP	C	0.045	0.000	0.000	0.043
G-2	SMAP	P	0.889	0.526	0.702	0.889
G-3	SMAP	P	0.533	0.000	0.990	0.533
G-4	SMAP	P	0.889	0.422	0.178	0.889
P-1	SMAP	C	0.169	0.000	0.718	0.240
P-2	SMAP	P	0.371	0.523	0.901	0.417
R-1	SMAP	P	0.667	0.022	0.022	0.667
T-2	SMAP	P	0.000	0.000	0.459	0.000
Mean			0.474	0.397	0.543	0.487

C | Per-Class and Per-Channel Sensitivity Results

This appendix disaggregates the aggregate sensitivity surfaces of Section 5.4 to the resolution at which the detector’s behaviour can be examined directly: by anomaly class, and by individual held-out channel. The data is the same controlled-injection campaign described in Section 5.4.1 — the nominally trained LSTM with its deployable $k = 3$ threshold, synthetic faults injected into clean nominal segments of the 20 held-out channels at the 6×5 grid of severity α and duration d , five replicates per cell — here decomposed rather than averaged over all channels. Channels are grouped by their primary class, so the one mixed channel (C-1) is counted with the point group, consistent with Chapter 5.

C.1. Detection performance by anomaly class

Tables C.1 and C.2 give the mean F_1 surface separately for the 14 point-anomaly channels and the 6 contextual-anomaly channels. The two surfaces explain the per-class contrast reported in Section 5.4.4. On point channels the surface is governed almost entirely by duration and is nearly flat in severity: at $d = 256$, raising severity from $\alpha = 0.10$ to $\alpha = 1.50$ changes the point-class F_1 by only $+0.08$ (from 0.731 to 0.811). On contextual channels severity matters strongly: the same increase raises the contextual-class F_1 by $+0.35$ (from 0.590 to 0.942), because a more severe contextual injection departs further from the nominal autocorrelation structure that the LSTM keys on. A subtle, short contextual fault ($\alpha = 0.10$, $d = 16$) is the single hardest case in the whole study, at $F_1 = 0.168$.

$\alpha \setminus d$	16	32	64	128	256
0.10	0.292	0.369	0.496	0.638	0.731
0.25	0.294	0.391	0.514	0.642	0.726
0.50	0.316	0.404	0.504	0.607	0.734
0.75	0.309	0.415	0.510	0.643	0.749
1.00	0.331	0.440	0.535	0.655	0.762
1.50	0.381	0.501	0.611	0.720	0.811

Table C.1: Point-anomaly channels ($n = 14$): mean F_1 across fault severity α and duration d (time steps), aggregated over the point channels of the held-out set.

$\alpha \setminus d$	16	32	64	128	256
0.10	0.168	0.258	0.442	0.555	0.590
0.25	0.235	0.317	0.541	0.683	0.788
0.50	0.357	0.440	0.685	0.825	0.949
0.75	0.434	0.554	0.685	0.803	0.917
1.00	0.478	0.618	0.722	0.831	0.915
1.50	0.581	0.701	0.799	0.892	0.942

Table C.2: Contextual-anomaly channels ($n = 6$): mean F_1 across fault severity α and duration d (time steps), aggregated over the contextual channels of the held-out set.

C.2. Detection latency by anomaly class

Tables C.3 and C.4 give the companion latency surfaces, in time steps from fault onset to first detection, averaged only over injections the detector actually flagged. Contextual faults are detected consistently later than point faults at equal severity and duration: a long, subtle contextual fault takes on the order of 40 to 50 steps to flag, against roughly 20 for the point equivalent. This is the temporal signature of the same effect seen in the F_1 surfaces — the LSTM accumulates the evidence for a contextual deviation more slowly — and it reinforces that the latency surface must be read jointly with the corresponding F_1 surface, never alone.

$\alpha \setminus d$	16	32	64	128	256
0.10	2.1	4.9	11.4	15.5	21.5
0.25	0.7	3.3	7.9	15.3	22.8
0.50	0.4	1.9	5.0	9.0	19.5
0.75	0.0	1.0	1.1	6.2	12.8
1.00	0.3	0.6	2.7	5.9	18.5
1.50	0.4	1.6	1.7	7.3	8.4

Table C.3: Point-anomaly channels ($n = 14$): mean detection latency (time steps to first detection) across severity and duration; lower is better.

$\alpha \setminus d$	16	32	64	128	256
0.10	6.7	9.3	16.8	22.1	46.5
0.25	5.6	8.7	26.9	29.1	51.2
0.50	4.2	5.5	12.1	26.2	44.2
0.75	2.7	5.9	8.9	22.6	38.3
1.00	2.4	4.0	4.6	12.9	24.0
1.50	1.6	2.4	4.1	6.6	15.2

Table C.4: Contextual-anomaly channels ($n = 6$): mean detection latency (time steps to first detection) across severity and duration; lower is better.

C.3. Per-channel detection at representative operating points

Table C.5 reports, for each of the 20 held-out channels, the detection F_1 at three representative operating points — a mild fault ($\alpha = 0.25$, $d = 64$), the operational anchor closest to a typical real fault ($\alpha = 1.00$, $d = 64$), and a severe, long fault ($\alpha = 1.50$, $d = 256$) — together with the no-injection false-positive rate of Section 5.4.5. The table exposes the dispersion that the aggregate surfaces average away. Several point channels (D-4, D-9, D-14, G-2) are detected reliably even at mild severity, while others (G-1, G-3) rise from near-zero detection at the mild point to above 0.84 when the fault is made severe and long — the channels for which severity, not merely duration, is decisive. Two channels,

D-5 and R-1, flag their entire clean nominal stream (a false-positive rate of 1.000) and correspondingly recover almost no usable detection; these two are what inflate the mean false-positive rate of Section 5.4.5 to 0.123 against a median of 0.011. Channel M-2 is the one channel whose clean segment is too short to host the longest injection, so its severe-fault entry is absent and it is the single channel excluded from the $d = 256$ column of the aggregate surface.

Channel	S/C	Type	Mild	Anchor	Severe	FP
			$\alpha 0.25$	$\alpha 1.0$	$\alpha 1.5$	
			d_{64}	d_{64}	d_{256}	
A-8	SMAP	C	0.397	0.870	0.996	0.000
B-1	SMAP	P	0.719	0.711	0.906	0.000
C-1	MSL	P	0.257	0.648	0.996	0.004
D-14	MSL	P	0.911	0.889	0.964	0.000
D-4	SMAP	P	0.946	0.932	0.982	0.003
D-5	SMAP	P	0.078	0.078	0.279	1.000
D-9	SMAP	P	0.899	0.863	0.978	0.000
E-10	SMAP	C	0.789	0.541	0.943	0.022
E-8	SMAP	P	0.329	0.325	0.683	0.153
F-3	SMAP	C	0.680	0.640	0.873	0.027
G-1	SMAP	C	0.100	0.902	0.993	0.000
G-2	SMAP	P	0.896	0.868	0.962	0.000
G-3	SMAP	P	0.000	0.000	0.840	0.000
G-4	SMAP	P	0.631	0.463	0.927	0.031
M-2	MSL	C	0.647	0.820	—	0.017
P-1	SMAP	C	0.635	0.557	0.906	0.056
P-2	SMAP	P	0.364	0.510	0.909	0.033
R-1	SMAP	P	0.083	0.083	0.295	1.000
T-2	SMAP	P	0.344	0.338	0.694	0.107
T-5	MSL	P	0.736	0.776	0.936	0.000

Table C.5: Per-channel detection F_1 at three representative operating points, with the no-injection false-positive rate (FP). Type is the channel’s primary class; C-1 is the mixed channel, grouped as point. A dash denotes an operating point the channel’s clean segment is too short to host.

D | Declaration on the Use of AI Tools

In the interest of transparency, this section declares the use of artificial-intelligence tools in the preparation of this thesis. The tools used were the large-language-model assistants ChatGPT (OpenAI) and Claude (Anthropic). Their use was discussed with my supervisor, and the assistance they provided is disclosed here rather than cited, as it does not constitute a citable source.

Scope. AI tools were used as an assistive aid for writing, programming and analysis. The research objectives, the experimental design, the choice and preparation of the dataset, the training and execution of every model and experiment, and the scientific conclusions are my own; AI was used to support and accelerate that work, not to replace the underlying research.

Text drafting and editing. AI tools were used to help draft and restructure prose, to improve clarity and English expression, and to prepare tables and figure captions. All text was reviewed, revised where necessary and approved by me, and reflects my own understanding of the work.

Code generation and debugging. AI tools assisted in writing and debugging the analysis, data-processing, statistical and plotting code. The implementation choices were mine, and I checked and validated the behaviour of the resulting code.

Result interpretation. AI tools were used as a discussion and sanity-checking aid when reasoning about results — for example, to help articulate findings, weigh alternative explanations and identify appropriate analyses. The final interpretation, the framing of every claim and the scientific conclusions are entirely my own.

List of Figures

- 2.1 SMAP channel P-1 telemetry with its three labelled contextual anomaly windows (shaded). The signal remains within its normal amplitude range of approximately $[-1, +1]$ throughout the anomaly windows, yet its oscillation pattern and statistical distribution change relative to the nominal behaviour learned during training. This is the category of fault that classical variance-based threshold detection cannot reach, and that motivates the distributional and predictive detectors developed in Chapter 3. 21
- 2.2 Closed-loop attitude-dynamics simulation of detumbling with a gyroscope drift fault injected at $t = 300$ s. The controller detumbles the spacecraft correctly during nominal operation; after fault onset the gyroscope estimate diverges from the true angular rate and the estimation error grows to approximately 14 deg/s. The running variance used by the classical FDIR monitor (following Colagrossi and Lavagna [3]) remains below its detection threshold throughout the fault period, so the fault is never flagged. Inertia values from Crotti and Colagrossi [4]; the drift rate and detection threshold are illustrative values chosen to expose the detection gap. 29
- 3.1 Supervised classification across the 81 unique channels. Per-channel F_1 for the SVM and the MLP, with contextual channels distinguished from point channels and the threshold-baseline mean shown for reference. The SVM achieves non-zero detection on every channel and a mean F_1 of 0.598. . . . 33
- 3.2 VAE anomaly detection under point-adjust across the 81 channels, with the published LSTM benchmark of Hundman et al. [15] shown for reference (SMAP 0.752, MSL 0.696). The VAE achieves a mean F_1 of 0.590 at the oracle threshold, with particular strength on contextual channels. 35
- 3.3 VAE detail for channel P-1. The reconstruction error is elevated within the three contextual anomaly windows but only moderately above the nominal training-error distribution, giving partial detection at the oracle threshold ($F_1 = 0.275$) and none at $k = 3$ 36

3.4	LSTM regression under point-adjust across the 81 channels. A majority of channels exceed both Hundman benchmark reference lines; the LSTM contextual-anomaly mean F_1 of 0.836 substantially exceeds both published values.	38
3.5	LSTM detail for channel P-1. The one-step-ahead prediction tracks the true signal during nominal operation and diverges inside all three contextual anomaly windows; the prediction error exceeds the $k = 3$ threshold, giving $F_1 = 0.726$ deployable and 0.814 at the oracle threshold, against the threshold baseline's 0.011.	39
4.1	Training curves for both generators (top: point class; bottom: contextual class). The gradient penalty stabilises below 0.05 by step 200 and the autocorrelation-matching loss falls below 0.003 within 100 steps and remains near zero, while the generator and critic losses reflect the cleaner convergence of the contextual distribution.	48
4.2	Real and synthetic anomaly windows for both classes. The synthetic contextual windows reproduce the oscillation structure of the real windows closely; the synthetic point windows capture the qualitative amplitude-excursion character with a residual gap on the most extreme excursions. . .	49
4.3	Two-dimensional t-SNE projection of real and synthetic windows. The synthetic contextual samples overlap the real contextual distribution closely; the synthetic point samples occupy a partially overlapping, more concentrated region, consistent with the heterogeneity of the real point distribution. .	51
5.1	Phase 4 on the 20 held-out channels. Left: synthetic-trained against real-trained deployable mean F_1 for the three detectors, with the held-out threshold baseline for reference; for the unsupervised detectors the deployable rule is the classical $k = 3$. Right: retention of the synthetic-trained detector relative to the like-for-like held-out real reference. The supervised SVM retains 75% with no statistically significant loss ($p = 0.13$); the unsupervised detectors fall back to the parameter-free $k = 3$ rule, which synthetic data does not improve.	57
5.2	Data-efficiency on the 20 held-out channels. The curve is flat: real anomaly windows from known channels do not transfer to improve detection on unseen channels. The Phase 2 all-channel baseline (0.598) is shown for reference only and is not directly comparable, because it is measured on a different and larger set of channels.	58

5.3	Augmentation on the 20 held-out channels under the clean protocol. Left (mean): synthetic-only training reaches the same plateau (about 0.47) as training on all cross-channel real windows, and adding real on top gives no further gain. Right (median): where cross-channel real data already yields a working detector, adding synthetic windows mildly degrades it, a weak residual distributional-drift signal. Synthetic substitutes for cross-channel real but does not augment it.	60
5.4	Detection-performance surface (mean and median F_1) over fault severity and duration, on the 20 held-out channels. Detectability increases monotonically with both axes; duration is the stronger determinant over the range examined. The $\alpha = 0$ column is the no-injection baseline of Section 5.4.5. This surface cannot be extracted from a fixed labelled dataset, in which each anomaly occurs at one uncontrollable operating point.	64
5.5	Detection-latency surface (mean time steps to first detection) over fault severity and duration, on the 20 held-out channels; lower is better. Latency is computed only over detected injections and must be read together with the F_1 surface of Figure 5.4.	66
5.6	Channel P-1 across all phases, as a held-out unseen-channel result. P-1 is a contextual anomaly that never exceeds the nominal amplitude bounds, so classical FDIR is blind to it ($F_1 = 0.011$). The nominally trained, $k = 3$ -thresholded LSTM recovers $F_1 = 0.718$ on this unseen channel, against the Phase 2 real-data LSTM oracle of 0.814.	68

List of Tables

1.1	Generative architectures reviewed in Section 1.3 and their role in this thesis.	16
2.1	NASA SMAP/MSL dataset statistics computed from <code>labeled_anomalies.csv</code> [15], over all 82 annotation rows (including the duplicated P-2). Aggregate detection results in later chapters use the 81 unique channels.	22
3.1	Detection performance across all 81 unique channels under point-adjust. [†] Published values from Hundman et al. [15], included for reference.	40
3.2	Mean F_1 disaggregated by anomaly class and spacecraft, under point-adjust.	40
3.3	Channel P-1 results across all methods. P-1 contains three contextual anomaly windows with no amplitude exceedance; the test-split anomaly rate is 8.8%.	41
4.1	Training configuration of the conditional WGAN-GP. Two generators are trained, one per anomaly class, under identical settings.	47
4.2	Generation-quality metrics by anomaly class for the clean generator (trained on the 61 training channels), computed against its real training windows; smaller is better. The contextual distribution is the more faithfully reproduced on the Fréchet metric, while both classes match the real autocorrelation structure closely.	50
5.1	Mean F_1 over the 20 held-out channels when real fault training data is replaced by synthetic data. The reference column is the same detector trained on real data and evaluated on the same channels. For the unsupervised detectors the deployable threshold is the classical $k = 3$ rule, which uses no fault data, so its value is unchanged by the source of the fault data.	56
5.2	Data-efficiency curve on the 20 held-out channels: mean and median F_1 as real anomaly windows from the training channels are added to synthetic training. The curve is flat.	58

5.3	Augmentation grid on the 20 held-out channels (supervised classifier, mean F_1): cross-channel real and synthetic windows reach the same plateau near 0.47, and adding synthetic windows on top of cross-channel real data gives no improvement on the mean. The “0 real, +0 synth” cell is omitted, as it leaves no training data.	59
5.4	Label-free threshold calibration on the 20 held-out channels (mean F_1). The oracle column requires the test labels and is an upper bound. * The apparent LSTM synthMedian win over $k = 3$ is a grid-resolution artefact, examined in Section 5.3.1.	61
5.5	Detection-performance surface: mean F_1 of the LSTM detector across fault severity α and duration d (time steps), aggregated over the 20 held-out channels (19 at $d = 256$).	64
5.6	Detection-latency surface: mean number of time steps from fault onset to first detection, across severity α and duration d , over the 20 held-out channels. The average at each cell is taken only over injections the detector actually flagged (Section 5.4.5).	65
5.7	Paired Wilcoxon signed-rank tests on the 20 held-out channels. “No significant difference” denotes failure to reject equality at the 0.05 level.	68
A.1	Supervised classifier (SVM and MLP) configuration.	82
A.2	Variational Autoencoder architecture and training configuration.	82
A.3	LSTM regression detector architecture and training configuration.	83
A.4	Generator architecture (per-class). Total parameters: 184,865.	84
A.5	Critic architecture. Spectral normalization (SN) is applied to every learnable layer; no batch normalization is used. Total parameters: 42,337.	84
B.1	Per-channel point-adjust F_1 for the Phase 1–2 detectors on all 81 unique SMAP/MSL channels. Channels held out from the generative experiments are marked †.	85
B.2	Per-channel point-adjust F_1 for the Phase 3–4 synthetic-to-real experiments on the 20 held-out channels. SVM (synth) is trained on synthetic data alone; VAE and LSTM are at the deployable $k = 3$ threshold.	88
C.1	Point-anomaly channels ($n = 14$): mean F_1 across fault severity α and duration d (time steps), aggregated over the point channels of the held-out set.	92

C.2	Contextual-anomaly channels ($n = 6$): mean F_1 across fault severity α and duration d (time steps), aggregated over the contextual channels of the held-out set.	92
C.3	Point-anomaly channels ($n = 14$): mean detection latency (time steps to first detection) across severity and duration; lower is better.	93
C.4	Contextual-anomaly channels ($n = 6$): mean detection latency (time steps to first detection) across severity and duration; lower is better.	93
C.5	Per-channel detection F_1 at three representative operating points, with the no-injection false-positive rate (FP). Type is the channel's primary class; C-1 is the mixed channel, grouped as point. A dash denotes an operating point the channel's clean segment is too short to host.	95

List of Symbols and Acronyms

Acronyms

ACF	Autocorrelation Function
AOCs	Attitude and Orbit Control System
AODS	Attitude and Orbit Determination System
DT	Digital Twin
ELBO	Evidence Lower Bound
ESA	European Space Agency
ESA-ADB	ESA Anomaly Detection Benchmark
FD	Fréchet Distance
FDIR	Fault Detection, Isolation and Recovery
GAN	Generative Adversarial Network
GNSS	Global Navigation Satellite System
GP	Gradient Penalty
KL	Kullback–Leibler (divergence)
KS	Kolmogorov–Smirnov (statistic)
LSTM	Long Short-Term Memory
MBSE	Model-Based Systems Engineering
MLP	Multilayer Perceptron
MMD	Maximum Mean Discrepancy
MSL	Mars Science Laboratory
PD	Proportional–Derivative (controller)
ReLU	Rectified Linear Unit
SMAP	Soil Moisture Active Passive

SVM	Support Vector Machine
TSTR	Train on Synthetic, Test on Real
TTUR	Two Time-scale Update Rule
VAE	Variational Autoencoder
WGAN-GP	Wasserstein GAN with Gradient Penalty
t-SNE	t-distributed Stochastic Neighbour Embedding

Symbols

Symbol	Description	Unit
F_1	Harmonic mean of precision and recall	—
α	Fault-injection severity (amplitude scale)	—
d	Fault-injection duration	time steps
k	Threshold standard-deviation multiplier	—
β	VAE Kullback–Leibler loss weight	—
λ_{GP}	Gradient-penalty loss weight	—
λ_{ACF}	Autocorrelation-matching loss weight	—
z	Latent / noise vector	—
μ, σ	Mean and standard deviation	—
θ_k	Detection threshold at multiplier k	—
ω	Angular-velocity vector	deg s^{-1}
I	Inertia tensor	kg m^2
τ	Control-torque vector	N m
τ	Fault onset time	s

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